Aga Khan University Examination Board

Notes from E-Marking Centre on HSSC-II Mathematics Examination May 2018

Introduction:

This document has been produced for the teachers and candidates of Higher Secondary School Certificate (HSSC) Part II (Class XII) Mathematics. It contains comments on candidates' responses to the 2018 HSSC - II Examination, indicating the quality of the responses and highlighting their relative strengths and weaknesses.

E-Marking Notes:

This includes overall comments on candidates' performance on every question and *some* specific examples of candidates' responses which support the mentioned comments. Please note that the descriptive comments represent an overall perception of the better and weaker responses as gathered from the e-marking session. However, the candidates' responses shared in this document represent some specific example(s) of the mentioned comments.

Teachers and candidates should be aware that examiners may ask questions that address the Student Learning Outcomes (SLOs) in a manner that require candidates to respond by integrating knowledge, understanding and application skills they have developed during the course of study. Candidates are advised to read and comprehend each question carefully before writing the response to fulfil the demand of the question.

Candidates need to be aware that the marks allocated to the questions are related to the answer space provided on the examination paper as a guide to the length of the required response. A longer response will not in itself lead to higher marks. Candidates need to be familiar with the command words in the SLOs which contain terms commonly used in examination questions. However, candidates should also be aware that not all questions will start with or contain one of the command words. Words such as 'how', 'why' or 'what' may also be used.

Key observations:

This year candidates did not perform well in questions based on applications of differentiation and integration and transformation. Candidates had problems with continuity of a function and finding equation of tangents from any point to circle. Candidates performed well on questions related to complex numbers, partial fractions, parabola, ellipse, and hyperbola.

Detailed Comments:

Constructed Response Questions (CRQs)

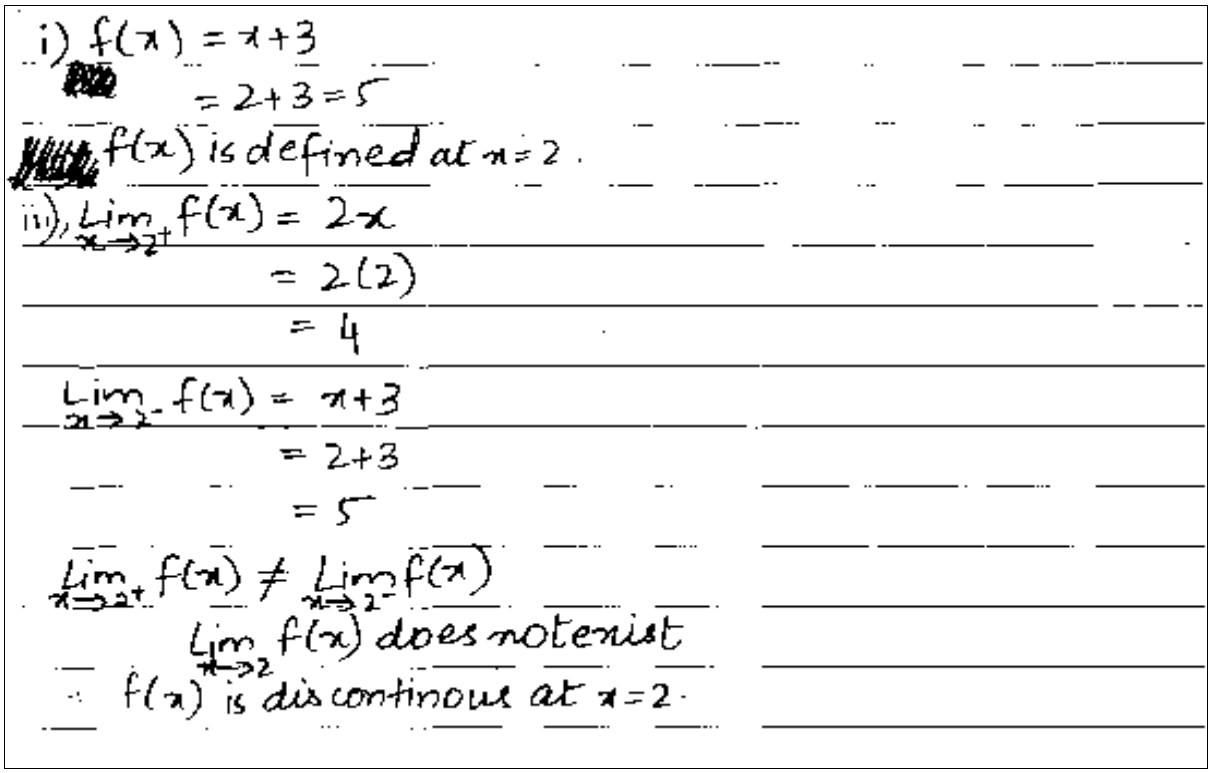
Question 1:

Determine whether the given function $f(x)$ is continuous at $x = 2$.

$$f(x)=\begin{cases} x+3 & \text{for } -2\le x\le 2 \\ 2x & \text{for } 2< x\le 4 \end{cases}$$

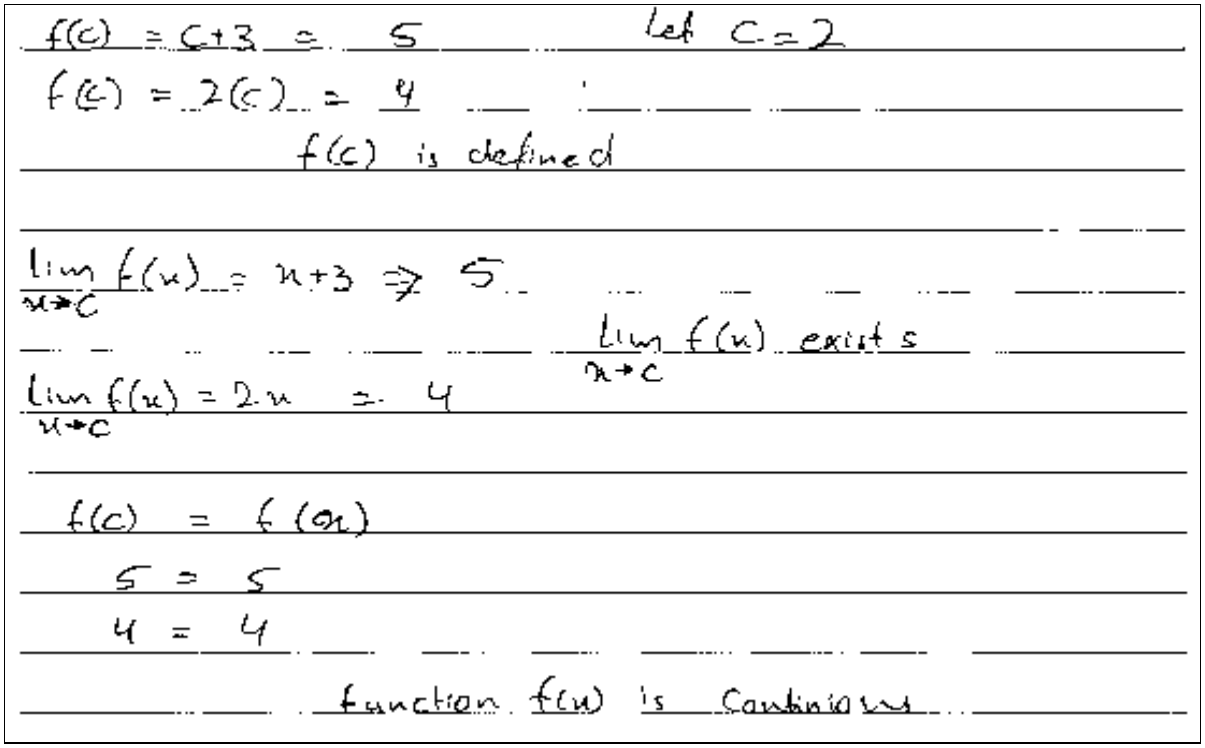
Better responses reflected that candidates understood the question very well as they knew the definition of continuity. This helped the candidates in applying all the conditions of continuity in order to conclude the desired result.

Example:



Weaker responses revealed vague understanding of continuity of a function. It was seen that candidates just substituted the values of x in the given function as they were unable to cater to all the conditions of continuity in order to conclude the required result. In some cases, candidates just found the limit for the function $f(x) = x + 3$ and got stuck in going further.

Example:



Question 2:

This question offered a choice to attempt any two questions from part **a**, **b** and **c**. Majority of the candidates attempted part **a** and **c** as compared to part **b**. It was evident that part **a** and **c** were attempted better than part **b**.

Question 2a:

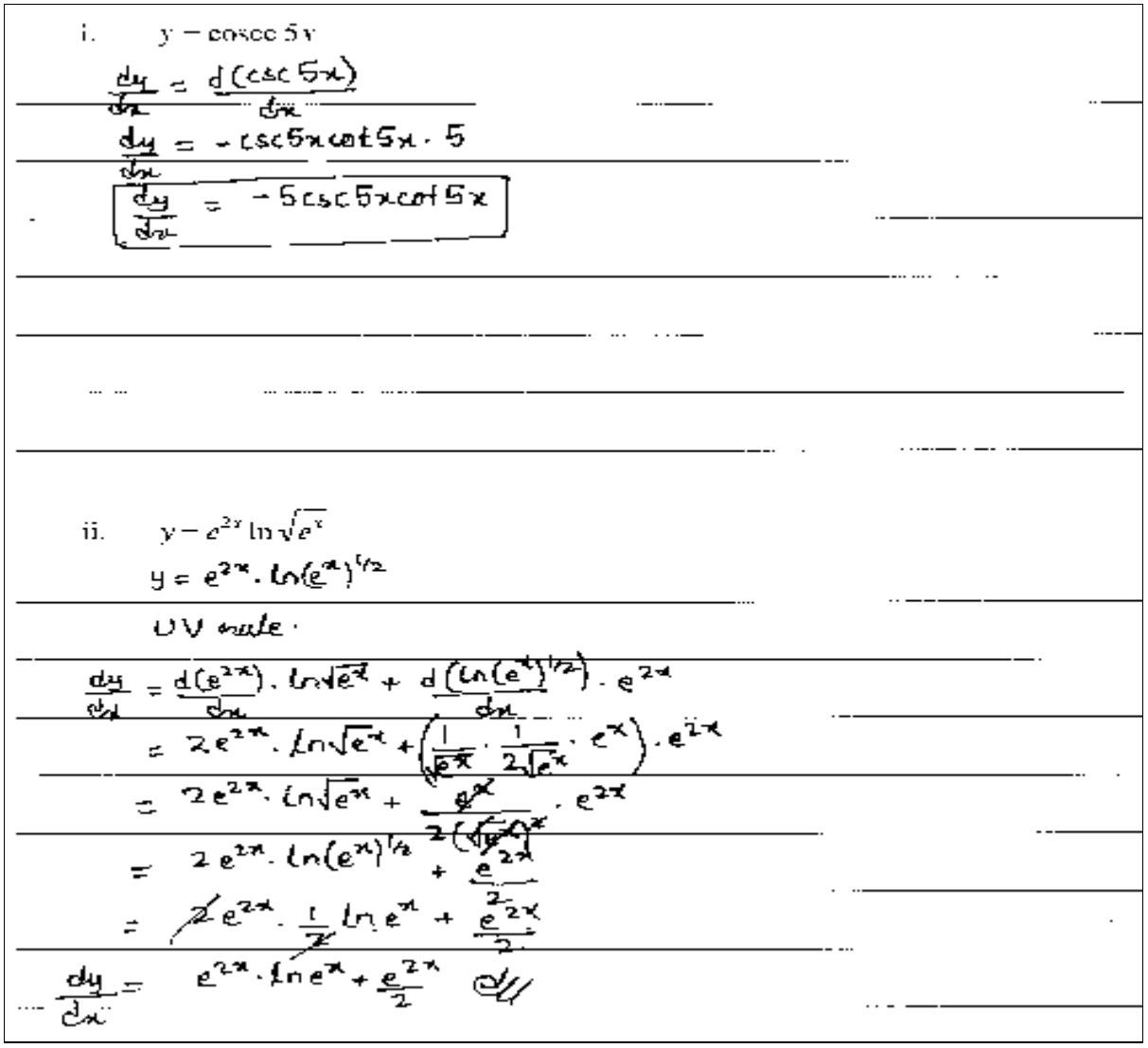
Differentiate the following functions with respect to x.

i. $y = \operatorname{cosec} 5x$

ii. $y = e^{2x} \ln \sqrt{e^x}$

Better responses showed the correct differentiation of trigonometric, exponential and natural logarithmic functions. This helped candidates in both parts to find the correct answer.

Example:



Weaker responses revealed that candidates' practice was not up to the mark as it should have been. In part i, most of the candidates failed to take the derivative completely, for example, they just wrote $\cos ec\ 5x \cot 5x$ rather than $5 \cos ec\ 5x \cot 5x$. It was also seen that they were even unable to take the derivative of $\cos ec\ 5x$. Similarly in part ii, they did not use the product rule and they just took the derivative of $e^{2x} \ln\sqrt{e^x}$ separately. For instance, they only managed to write $2e^{2x} \frac{1}{\sqrt{e^x}}$ which led to an incomplete and incorrect answer.

Example:

i. $y = \operatorname{cosec} 5x$

$$y = \frac{1}{\sec} \cdot 5x$$

$$u \cdot v = uv' + vu'$$

$$= 5x \cdot \csc\theta + 5 \cdot \frac{1}{\sec\theta}$$

$$= \frac{5x \cdot 1}{\sec\theta \sec\theta} + \frac{5}{\sec\theta}$$

$$= \frac{5x + 5}{\sec x} = \frac{5(x+1)}{\sec x}$$

ii. $y = e^{2x} \ln\sqrt{e^x}$

$$y = e^{2x} \ln(e^x)^{\frac{1}{2}}$$

$$y = e^{2x} \cdot \ln(e^{\frac{x}{2}})$$

$$u \cdot v = uv' + vu'$$

$$= e^{2x} \cdot \frac{1}{x} (e)^{\frac{x}{2}}$$

$$= e^{2x} \cdot \frac{x}{2x} (e)$$

$$= \frac{1 \cdot e^{2x} \cdot e}{2}$$

$$= \frac{1}{2} (e)^{2x}$$

Question 2b:

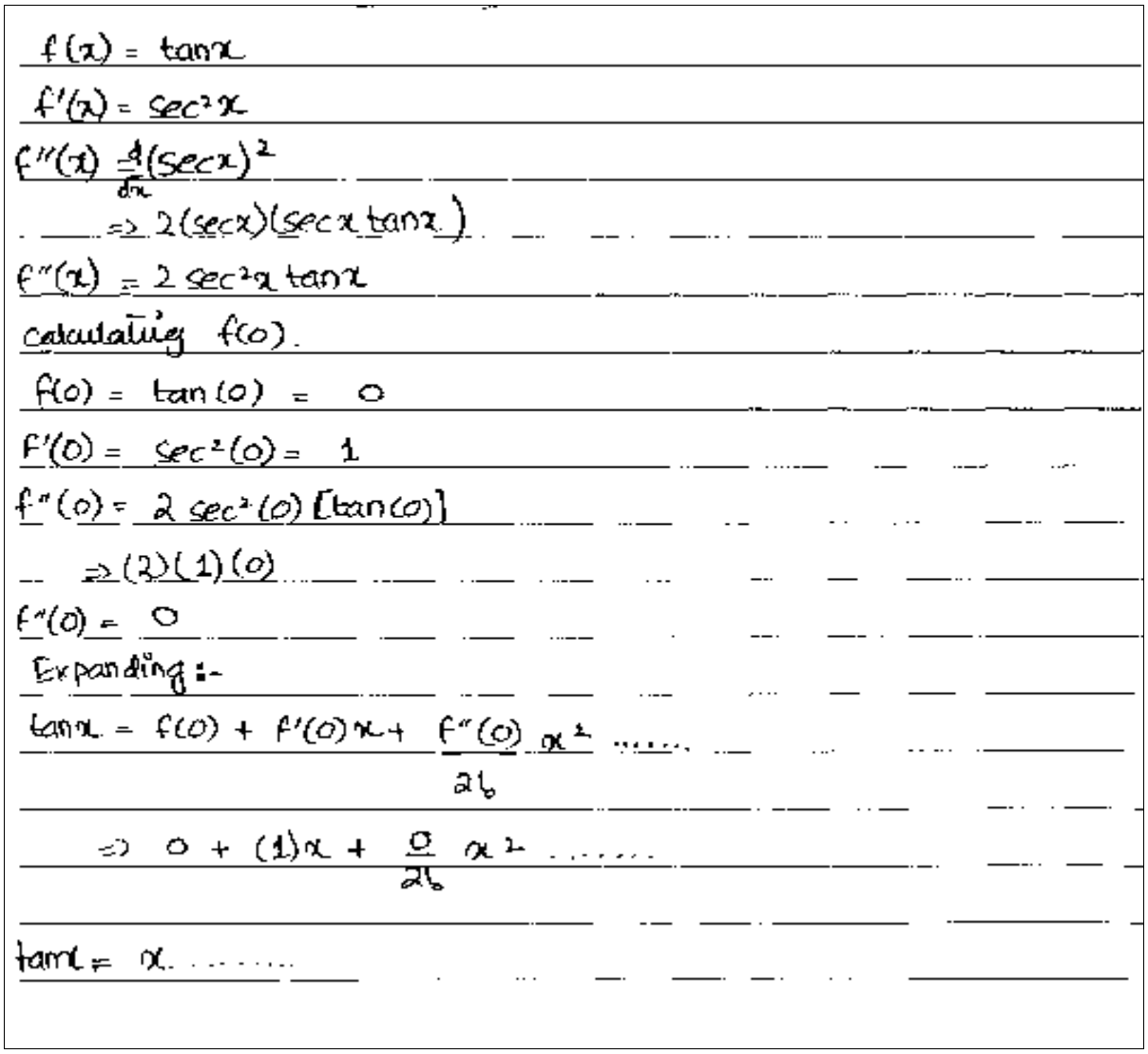
Expand the Maclaurin series for $\tan x$ up to 3rd term. Show your working.

The Maclaurin expansion for $f(x)$ about $x = 0$ is given as

$$f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \ldots$$

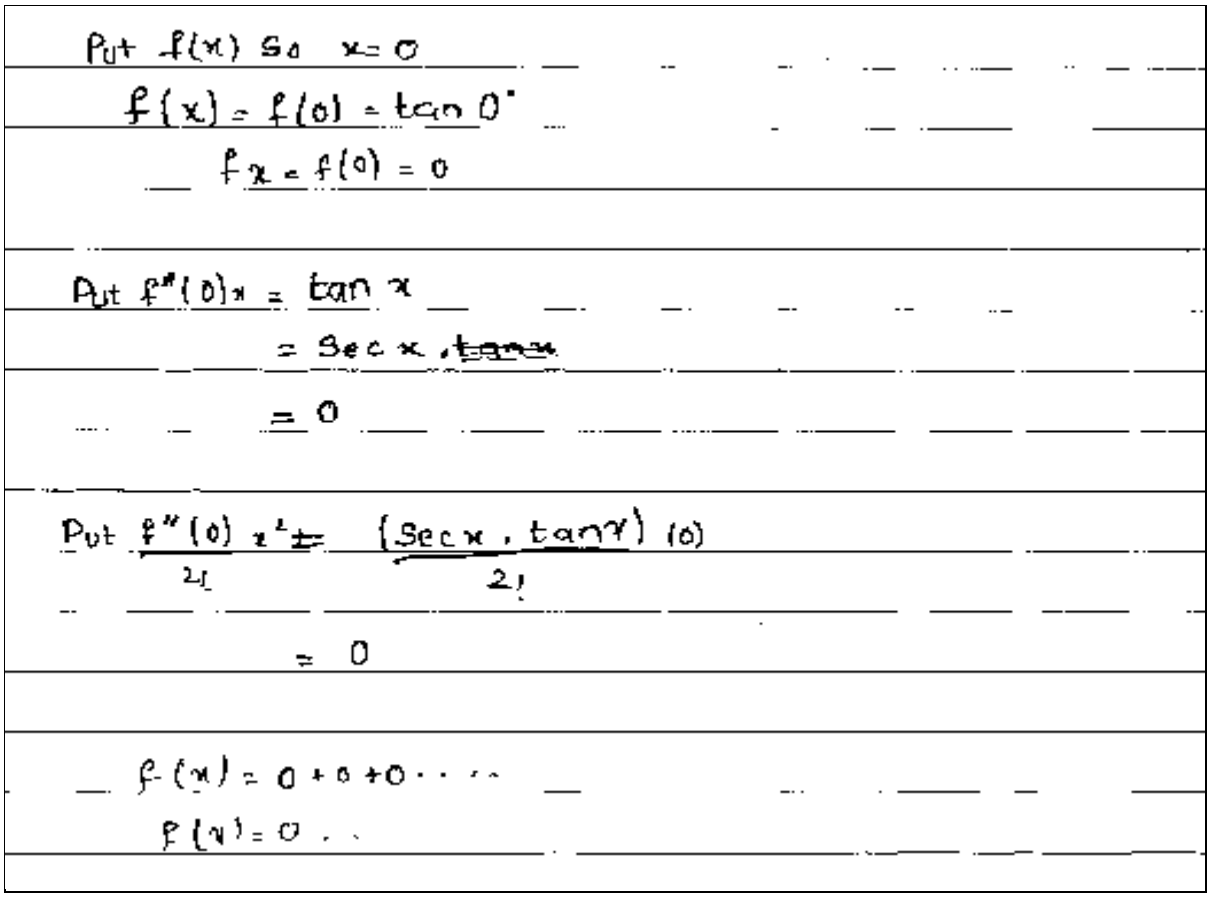
Better responses demonstrated that candidates used the Maclaurin's series for expanding the function $f(x) = \tan x$ very well. They took the derivative and substituted the values accordingly which resulted in the correct expansion of the expression.

Example:



Weaker responses revealed that candidates were not fully aware of Maclaurin's series for expanding the function $f(x) = \tan x$. They just substituted $x = 0$ in $f(x)$ and failed to take derivative and hence, couldn't move further. As a result, they were unable to expand completely.

Example:



Question 2c:

The volume of a paper box is expressed as $V = 27x - \frac{x^3}{4}$, where the length of a side of the box is x cm.

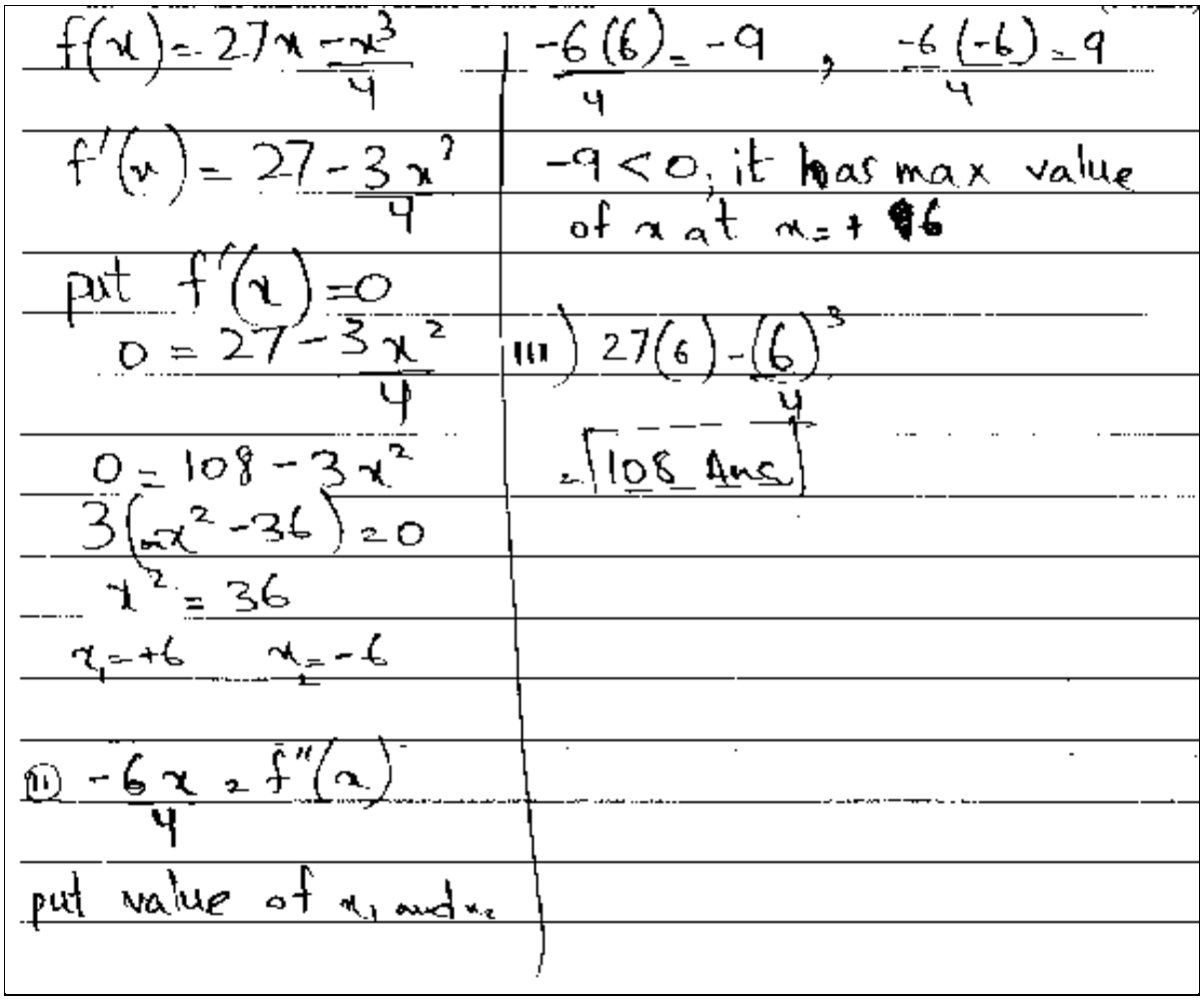
i. Find the value of x for which the volume of the box is maximum.

ii. Show that the volume of the box at x is maximum.

iii. Find the maximum volume of this box.

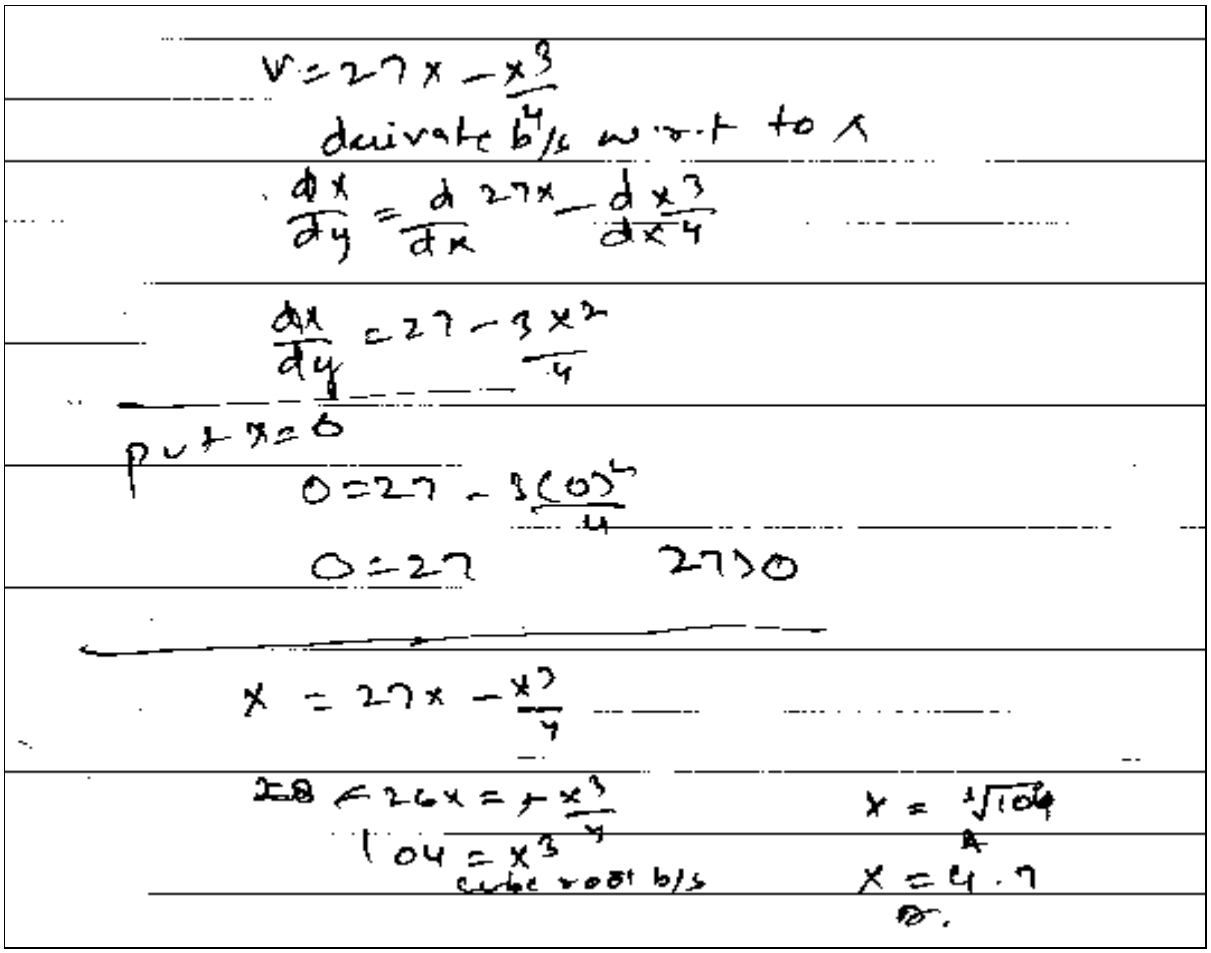
Better responses showed the correct application of differentiation which highlighted the candidates' ability to get the maximum volume. The first and second derivative tests were used to find extreme points and their nature. Hence, the maximum value of the function was found.

Example:



Weaker responses reflected the confusion of candidates after finding the first derivative, put it equal to zero, and got two values of x i.e., -6 and 6. They had chosen $x = -6$ which was actually wrong because, sides can never be negative and for this value of x they decided that the function is maximum rather than going for the second derivative test. It was also seen that fewer candidates even could not find the derivative correctly.

Example:



Question 3:

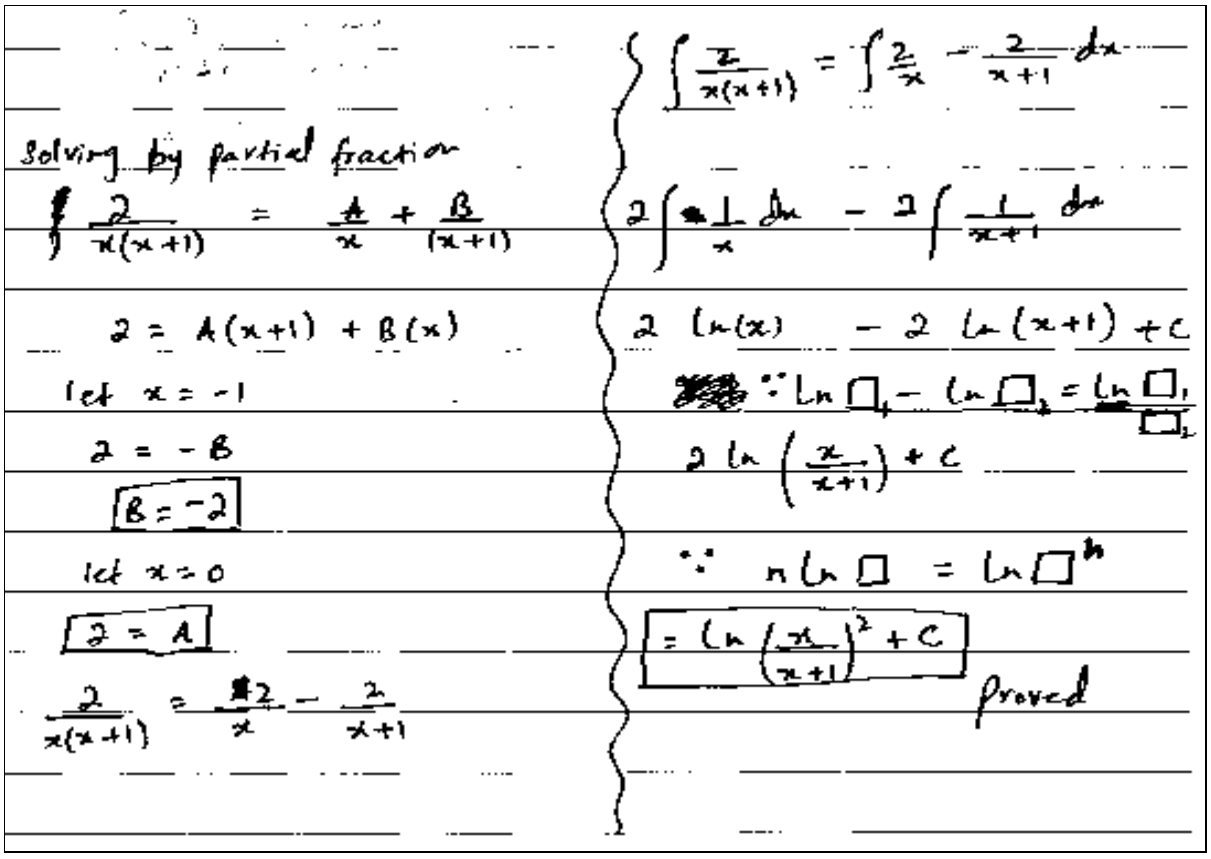
This question offered a choice to attempt any two questions from parts **a**, **b** and **c**. Majority of the candidates attempted part **a** and **c**. All parts were well attempted.

Question 3a:

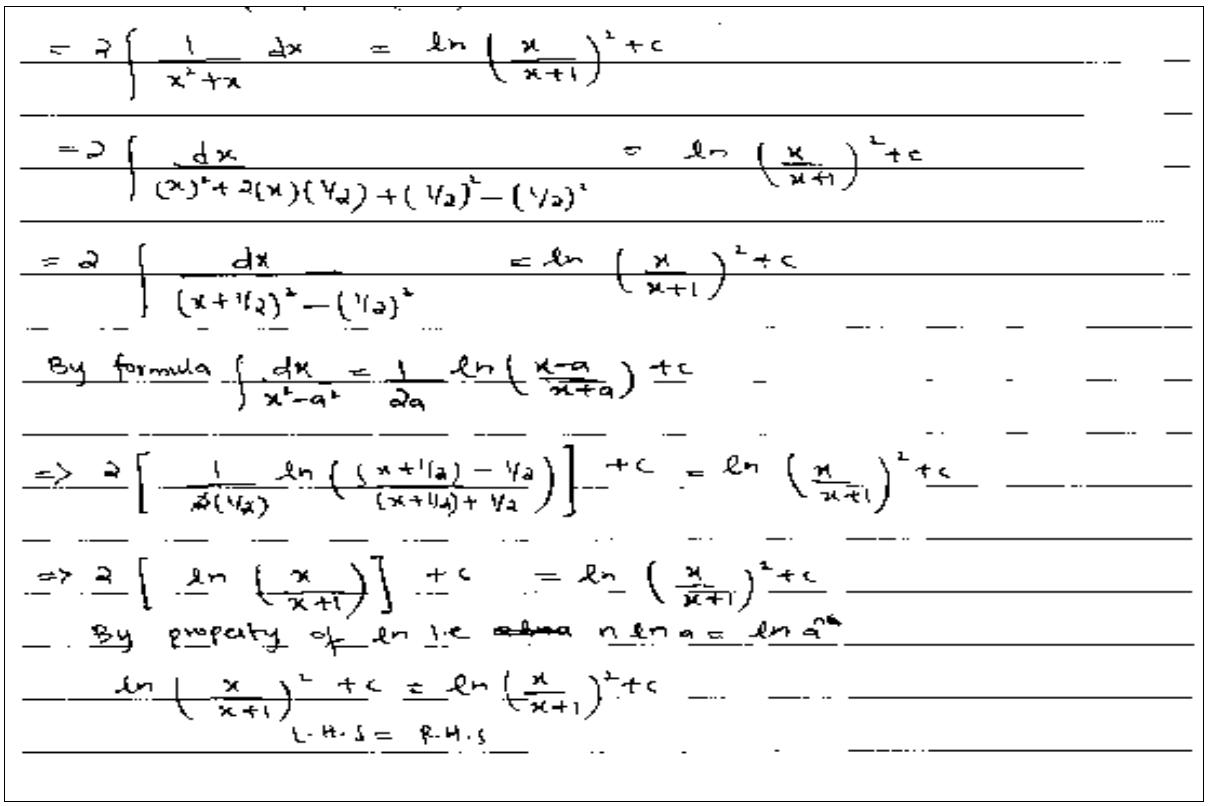
Show that $\int \frac{2}{x(x+1)}dx = \ln\left(\frac{x}{x+1}\right)^2 + c$.

Better responses well understood the question and used partial fraction to integrate in order to reach the required form. Multiple responses were seen which reflected the beauty of Mathematics and candidates' skills.

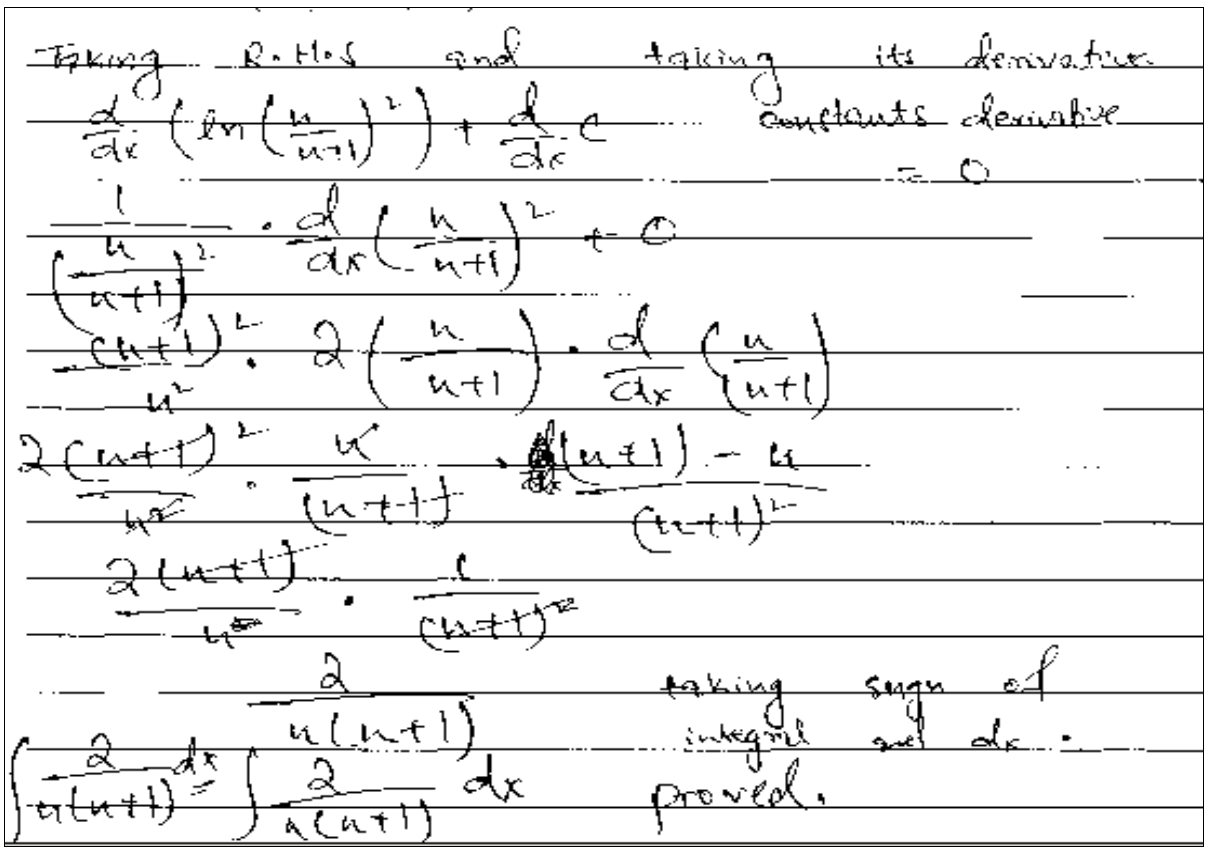
Example 1:



Example 2:

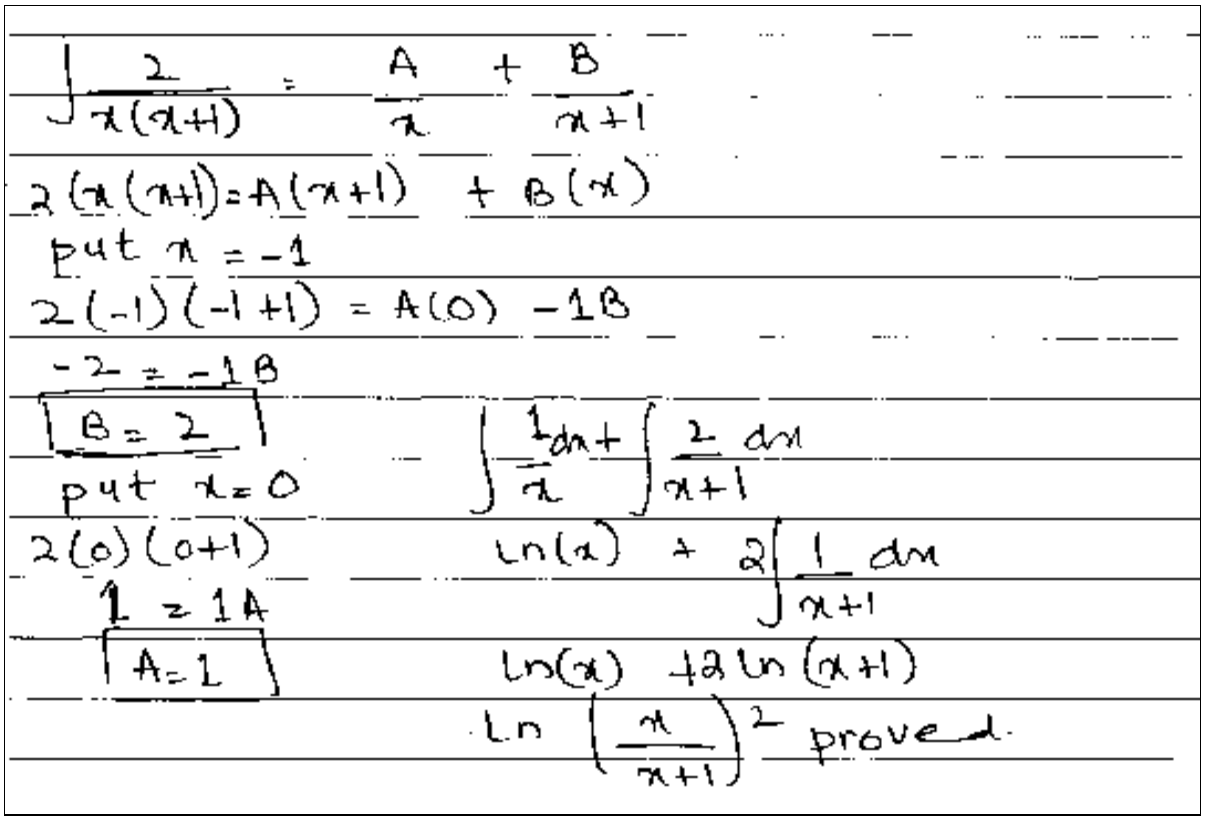


Example 3:



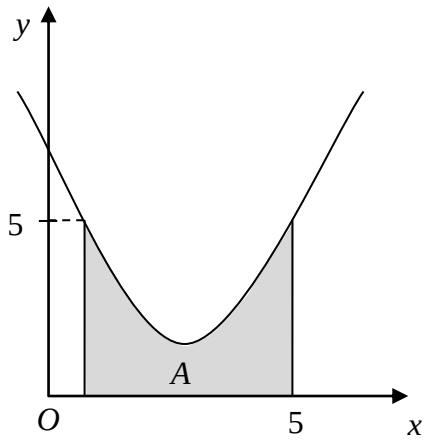
Weaker responses reflected that candidates had faced difficulty in understanding the question. They were unable to convert the fractions in order to integrate and got stuck in finding the required form.

Example:



Question 3b:

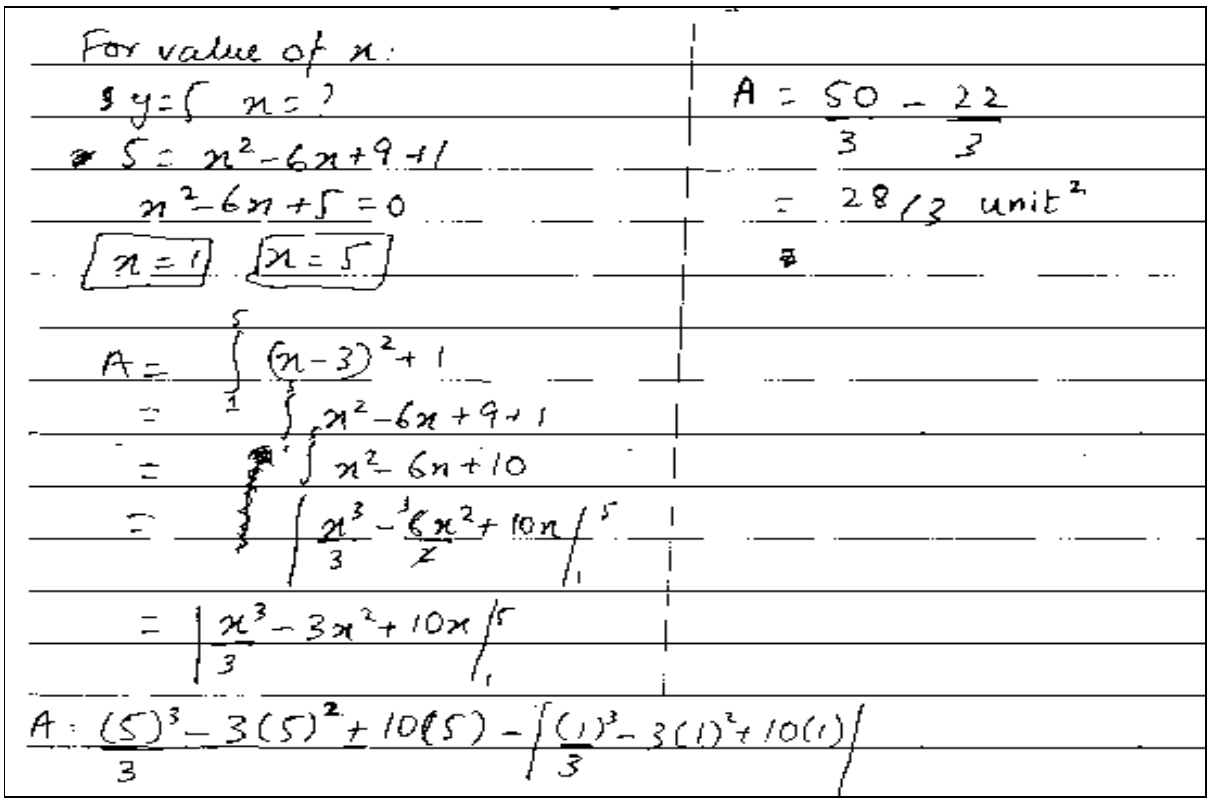
The given diagram shows the curve of $y = (x-3)^2 + 1$. Find the area of the shaded region *A*.



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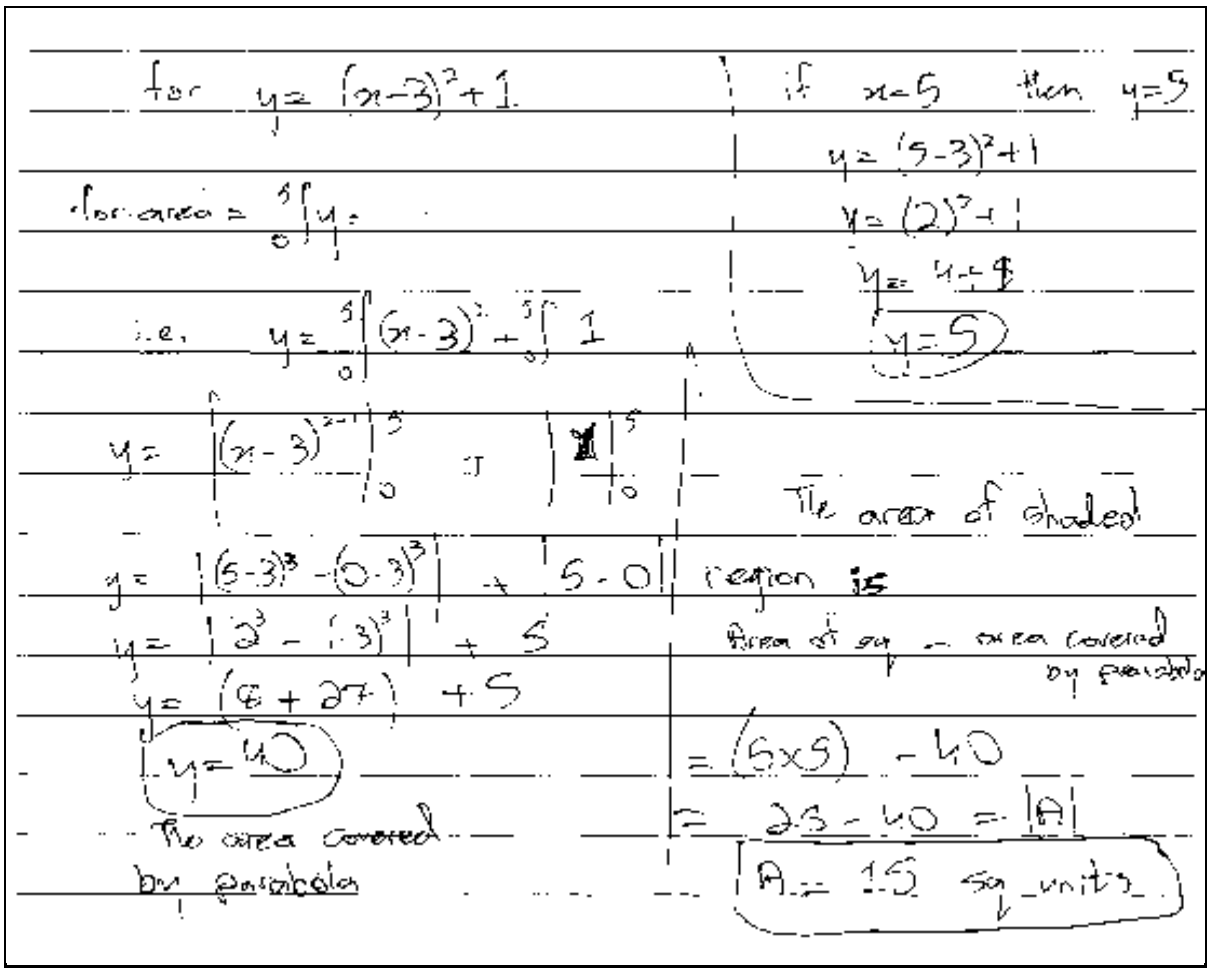
Better responses showed that candidates integrated the equation of curve as $y=(x-3)^2+1$ to find the area of region *A*. They found the values of *x* followed by the correct integration which led them to the required answer. It was also seen that few candidates directly integrated under the limit 1 till 5 without finding one value of *x*.

Example:



Weaker responses exhibited that candidates failed to integrate even though they found the values of *x*. Because of incorrect integration, they were unable to reach the required answer.

Example:

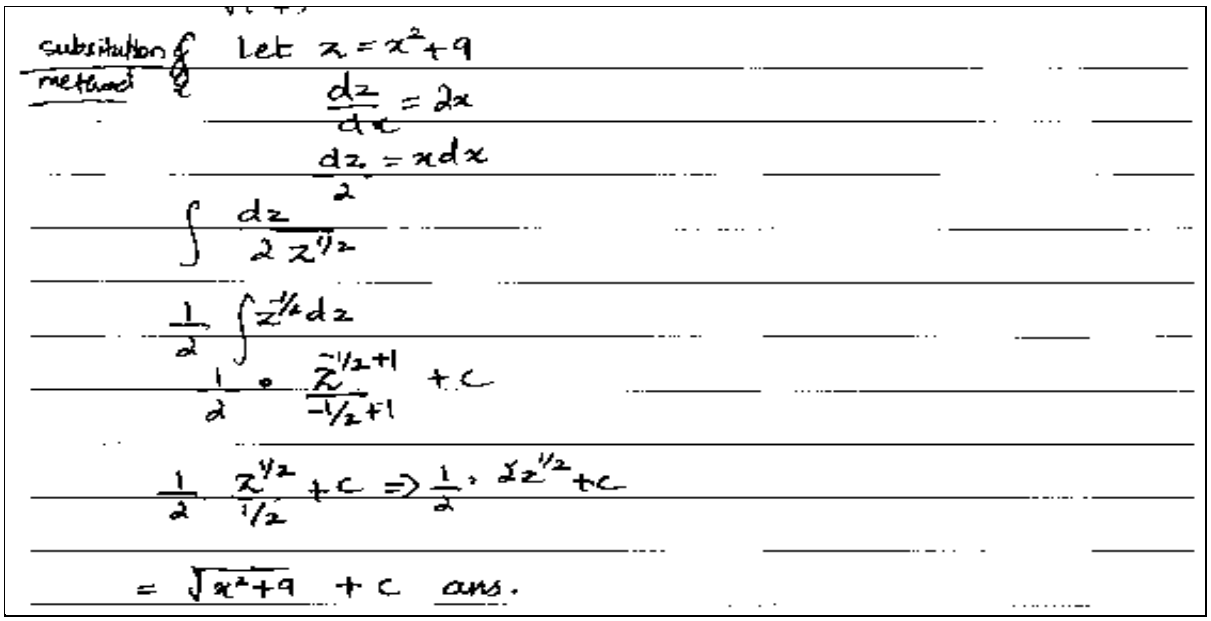


Question 3c:

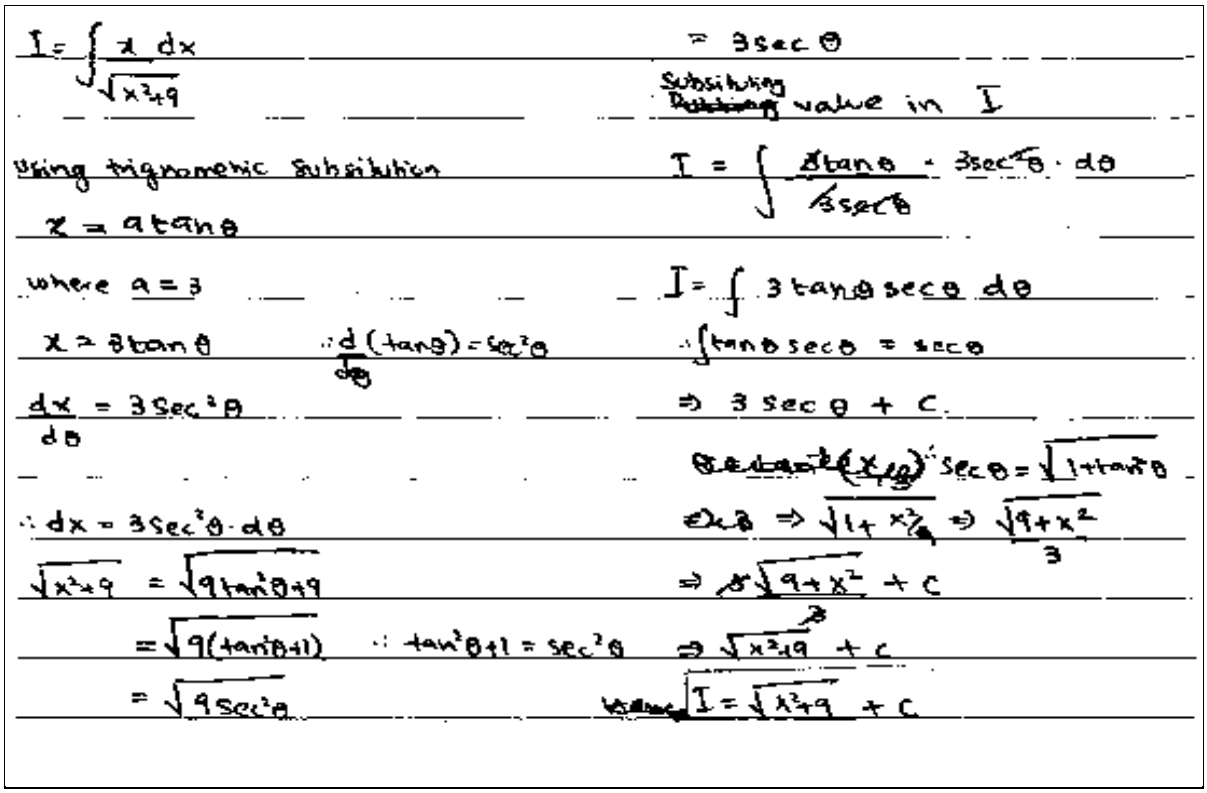
Evaluate $\int \frac{xdx}{\sqrt{x^2+9}}$.

Better responses indicated that most of the candidates were able to integrate by using substitution or by using derivative. This helped the candidates to integrate easily and hence, the final answer was evaluated. Different responses were observed as well.

Example 1:

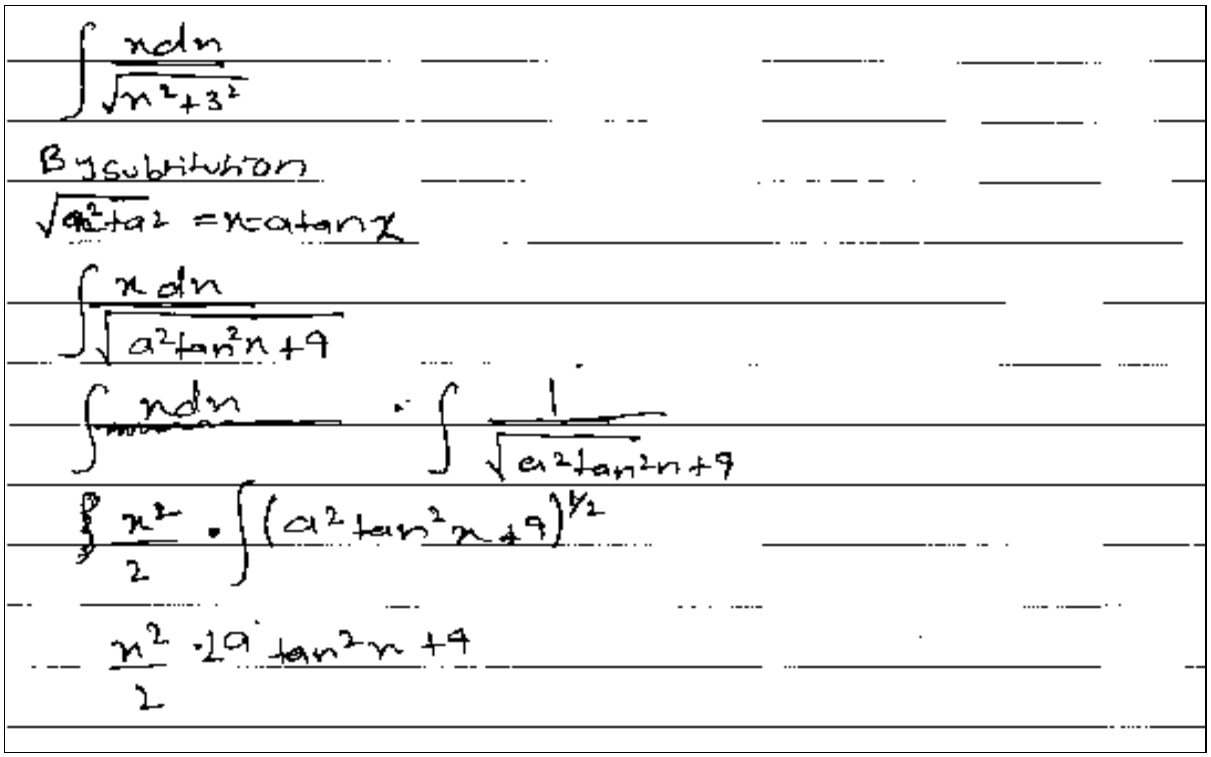


Example 2:



Weaker responses reflected that candidates did not understand the question and it was seemed to be a lack of practice. Candidates were unable to go further although they used the correct substitution.

Example:



Question 4:

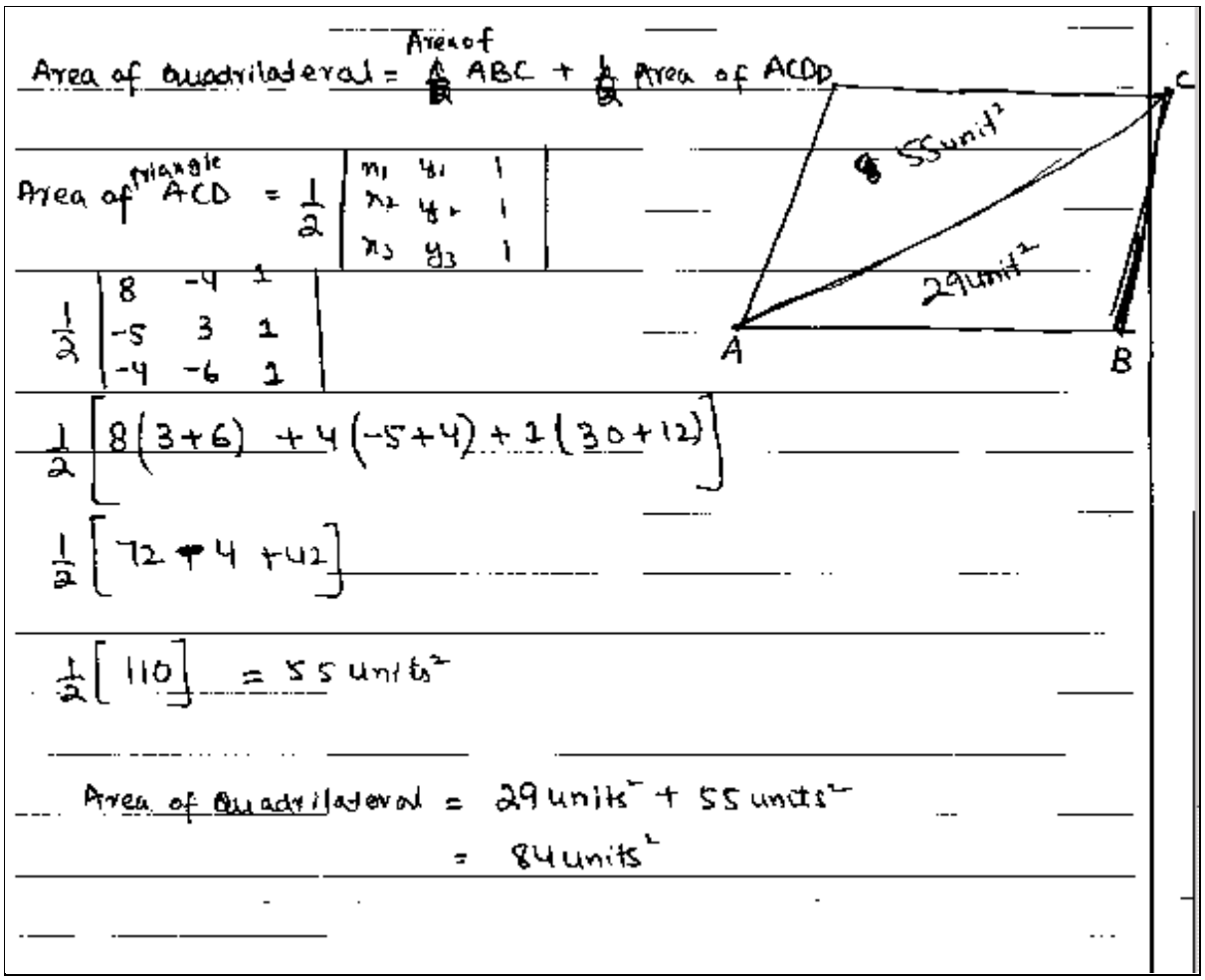
This question offered a choice to attempt any two questions from part **a**, **b** and **c**. Majority of the candidates attempted part **a** and part **b**. Also part cii was well attempted.

Question 4a:

A quadrilateral *ABCD* has vertices *A* (8, – 4), *B* (7, 1), *C* (– 5, 3) and *D* (− 4, − 6). If the area of triangle *ABC* is 29 units^2, then find the area of the quadrilateral *ABCD*.

Better responses exhibited that candidates were able to understand the question and used the data accordingly. This helped candidates in applying the formula of area of triangle to get the correct answer.

Example:



Weaker responses reflected that most of the candidates failed to write the coordinates in order although it was mentioned in the question which showed the lack of knowledge. As a result, they could not get the right path in order to get the answer.

Example:

If we suppose quadrilateral in two triangle



for Area of ABC:-

$$A = \frac{1}{2}\begin{vmatrix} 8 & -4 & 1 \\ 7 & 1 & 1 \\ -5 & 3 & 1 \end{vmatrix}$$

$$A = \frac{1}{2}(58)$$

$$\boxed{ABC = 29 \text{ unit}^2}$$

now for triangle BCD:- B(7,1)(-5,3)(-4,-6)

$$A :- \frac{1}{2}\begin{vmatrix} 7 & 1 & 1 \\ -5 & 3 & 1 \\ -4 & -6 & 1 \end{vmatrix}$$

$$A :- \frac{1}{2}(106)$$

Area of BCD = 53 unit²

Total Area of Q:-

Area of ABC + Area of BCD

= 29 + 53

$$= \boxed{82 \text{ unit}^2}$$

Question 4b:

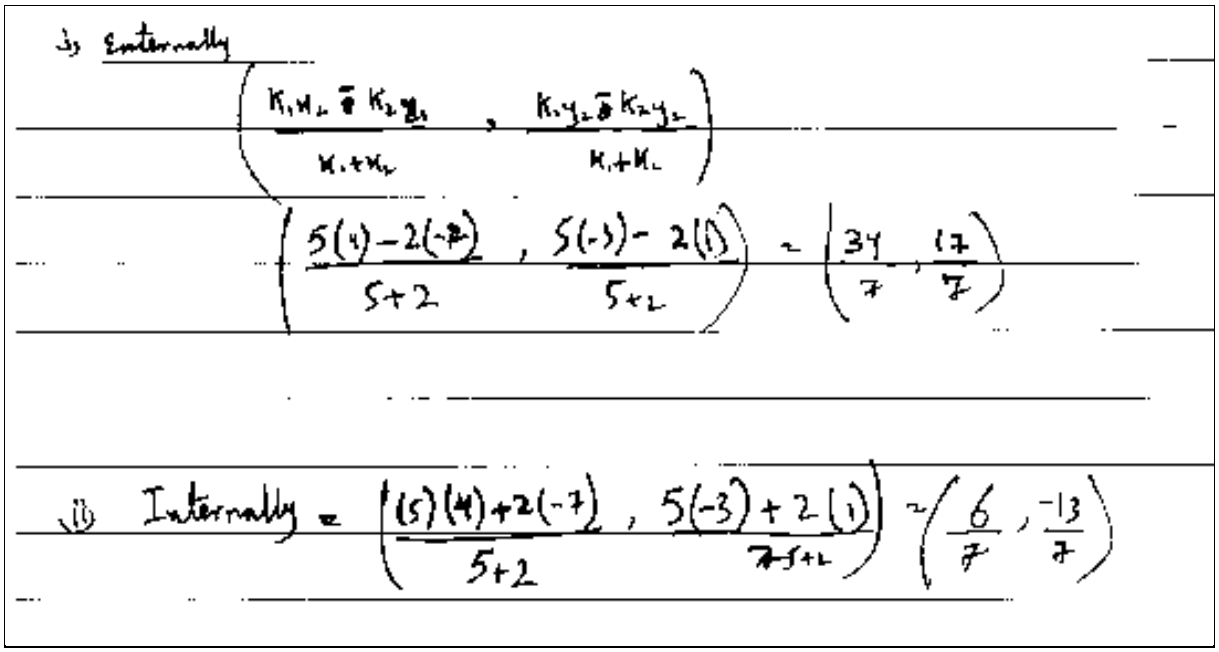
A line segment is formed by joining the points $P(-7, 1)$ and $Q(4, -3)$. Find the coordinates of a point Z that divides this line in the ratio 5 : 2

i. externally.

ii. internally.

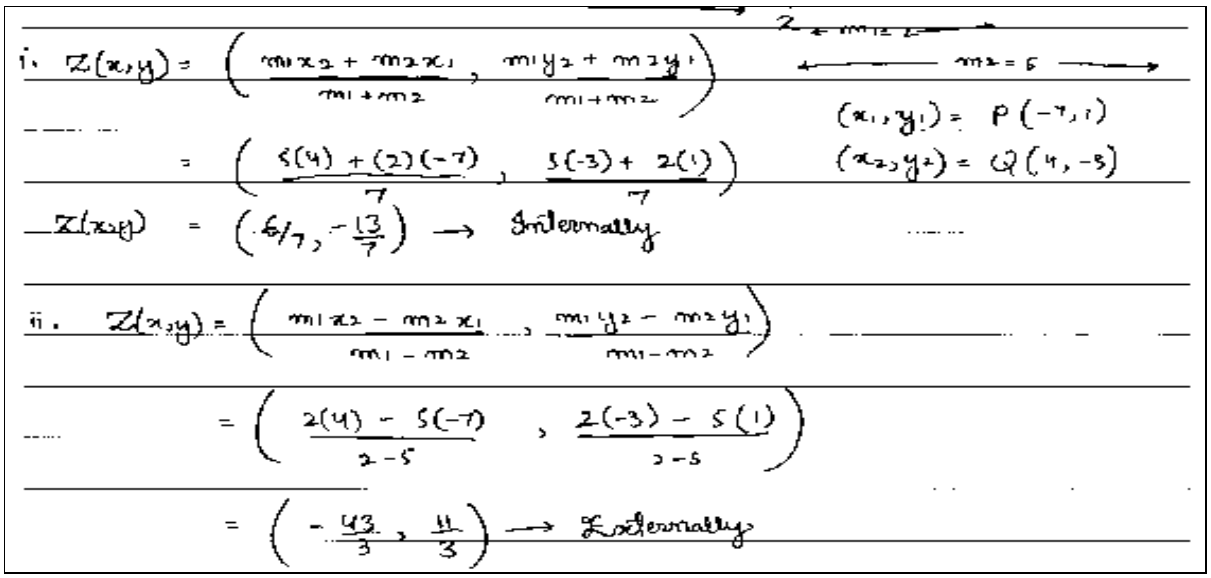
Better responses reflected correct application of formula and used the data in order to substitute the values accordingly. This helped candidates to reach the required answer successfully.

Example:



Weaker responses exhibited that the candidates failed to write the formula of internal and external division correctly. Although question was easy but, due to lack of practice they were unable to reach the required answer. It was also seen that candidates got perplexed and mixed up the formula for internal and external division with one another which resulted in incorrect answers.

Example:

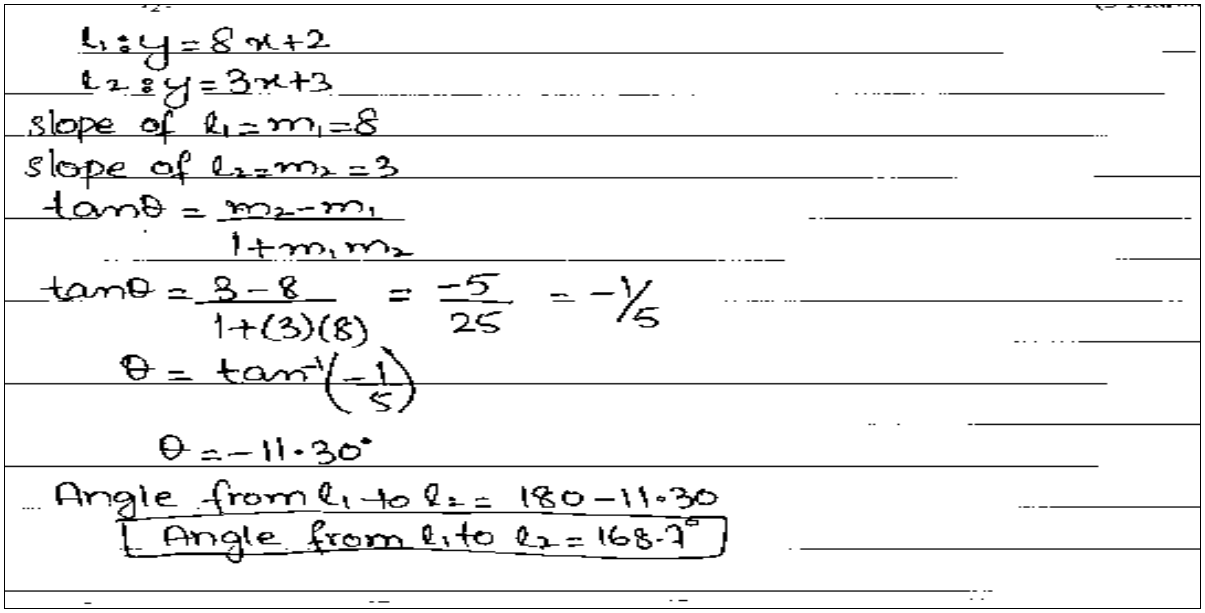


Question 4c:

i. The lines l_1, l_2 and l_3 are the sides of a triangle. The lines l_1 and l_2 are given as $l_1 : y = 8x + 2$ and $l_2 : y = 3x + 3$. Find the angle of the triangle that is measured from l_1 to l_2.

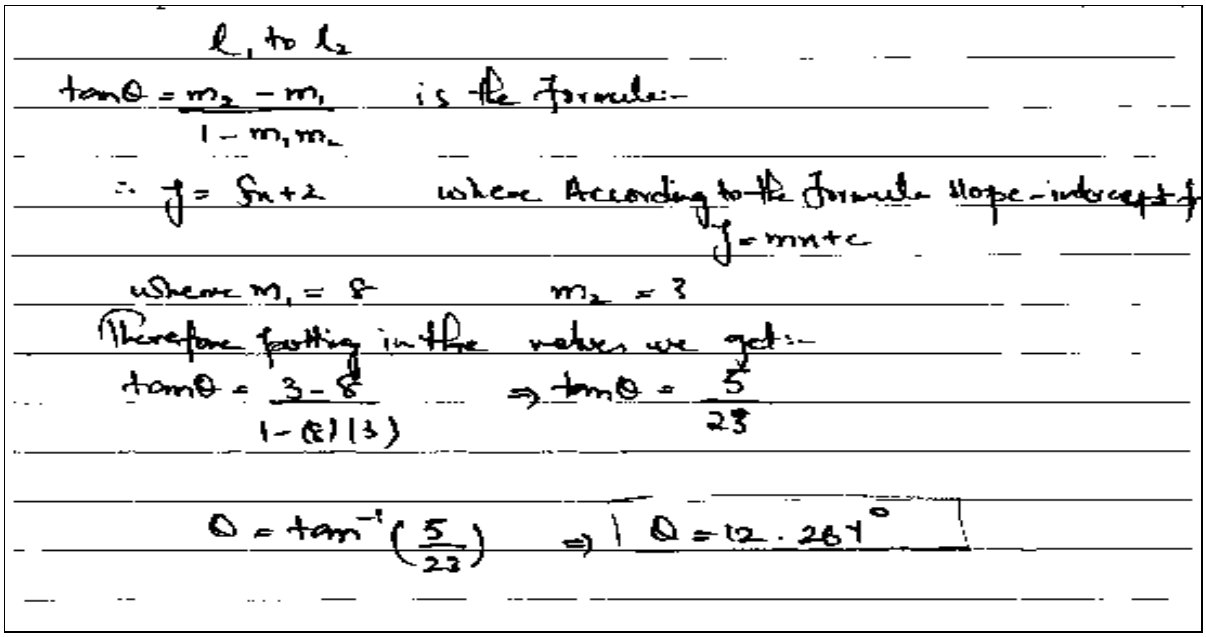
Better responses reflected correct application of formula and substitution of values m_1 and m_2. The angle from l_1 to l_2 was found, i.e., 168.7°, by finding basic angle using $\tan\theta = -\frac{1}{5}$. Majority of the candidates, however, did not find the basic angle. Therefore, the most frequent answer found by candidates was – 11.3° which is the clockwise angle instead of the anticlockwise answer required.

Example:



Weaker responses exhibited incorrect substitution of m_1 and m_2 in the formula. After substitution, minor arithmetic errors were frequent too. As a result, they could not reach the required answer.

Example:

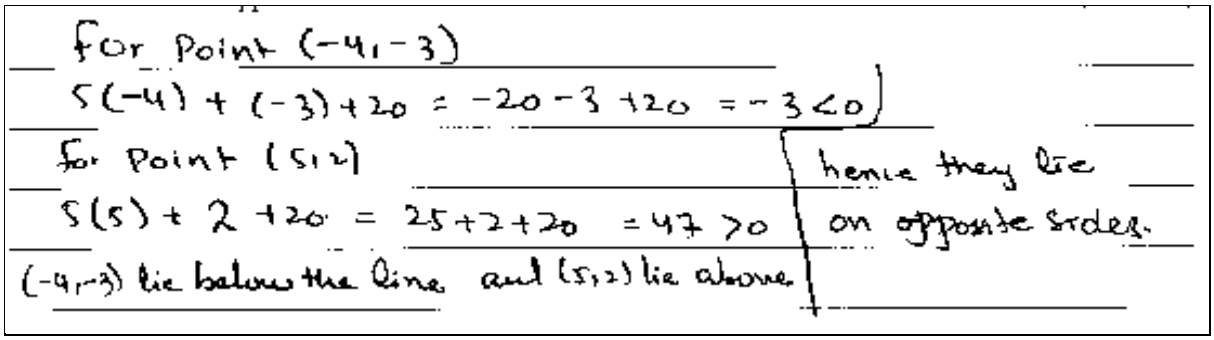


Question 4c:

ii. For the line $5x + y + 20 = 0$, determine if the points $(-4, -3)$ and $(5, 2)$ lie on the same side or opposite sides.

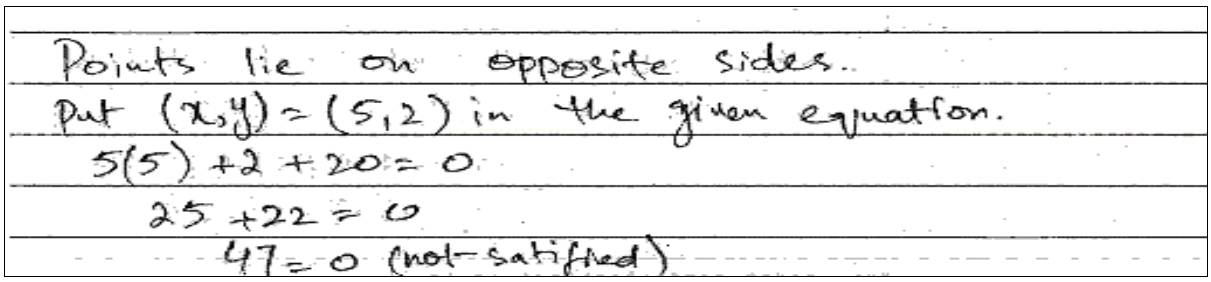
Better responses demonstrated that candidates had a good understanding of sign significance for the given equation of line.

Example:



Weaker responses demonstrated that candidates did not understand the question as they were unable to check the sign significance for the given equation of line.

Example:



Question 5:

This was not a well attempted question.

Question 5:

A furniture company produces tables and chairs. The production process consists of carpentry and painting. Each table takes 5 hours in carpentry and 3 hours in painting department. Each chair requires 3 hours in carpentry and 1 hour in painting department. During a certain production period, 360 hours of carpentry time and 250 hours of painting time are available. Each table sold yields a profit of Rs 1,500 and each chair sold yields a profit of Rs 800.

Production Process	**Hours Required to make 1 Unit**		**Available Hours**
	Table	**Chair**	
Carpentry	5	3	360
Painting	3	1	250
Profit	1,500	800	

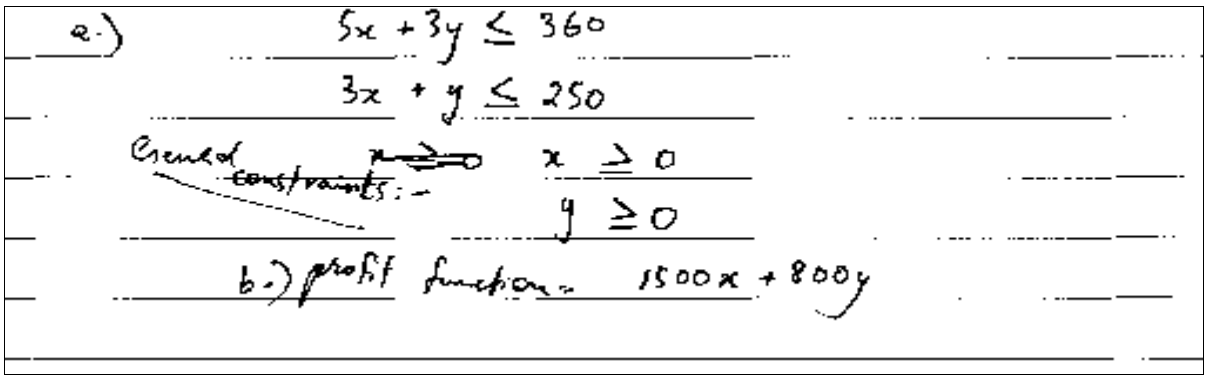
Suppose that *x* is the number of tables and *y* is the number of chairs produced in this production period, formulate this problem as a linear programming problem by finding the

a. constraints.

b. profit function.

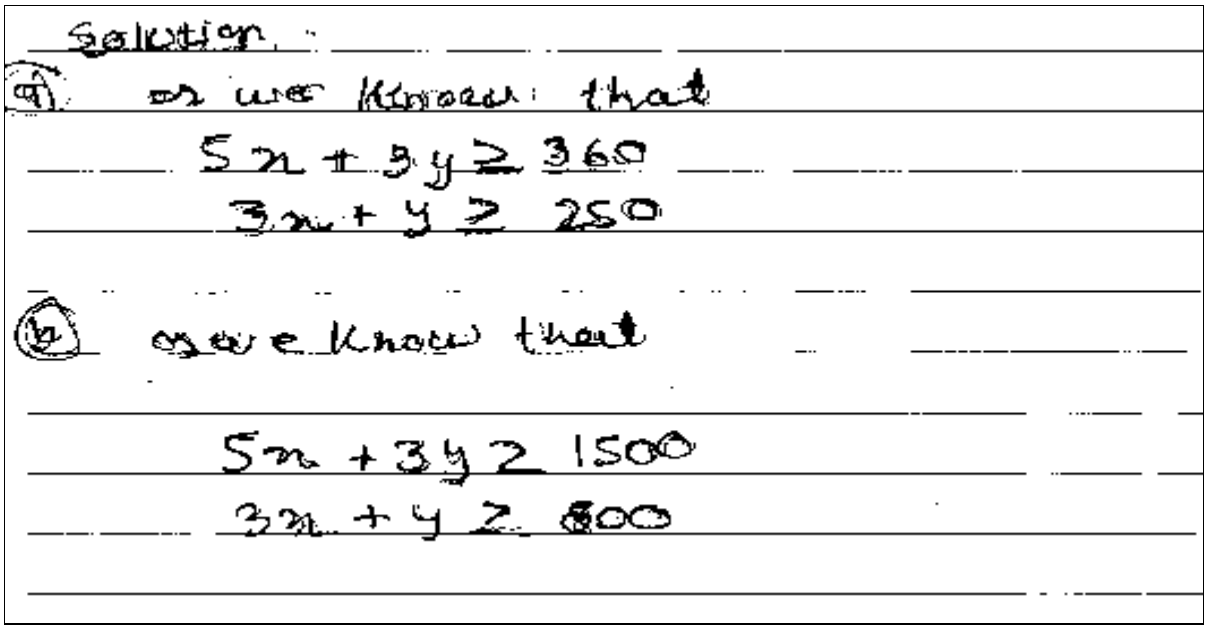
Better responses revealed that the candidates did well to understand the given situation. The given variables *x* and *y* in the question were correctly used, along with the values given in the table, to construct meaningful constraints and cost functions.

Example:



Weaker responses reflected weak understanding of application of linear programming with most of the students giving constrains and cost functions that made no mathematical sense according to the given situation. It is generally a difficult topic for candidates to facilitate them. It was also noticed that fewer candidates got perplexed in using the inequality sign.

Example:



Question 6:

This question offered a choice between part **a** and part **b**. Majority of the students chose part **b** as compared to part **a**. Not many candidates were able to score full marks.

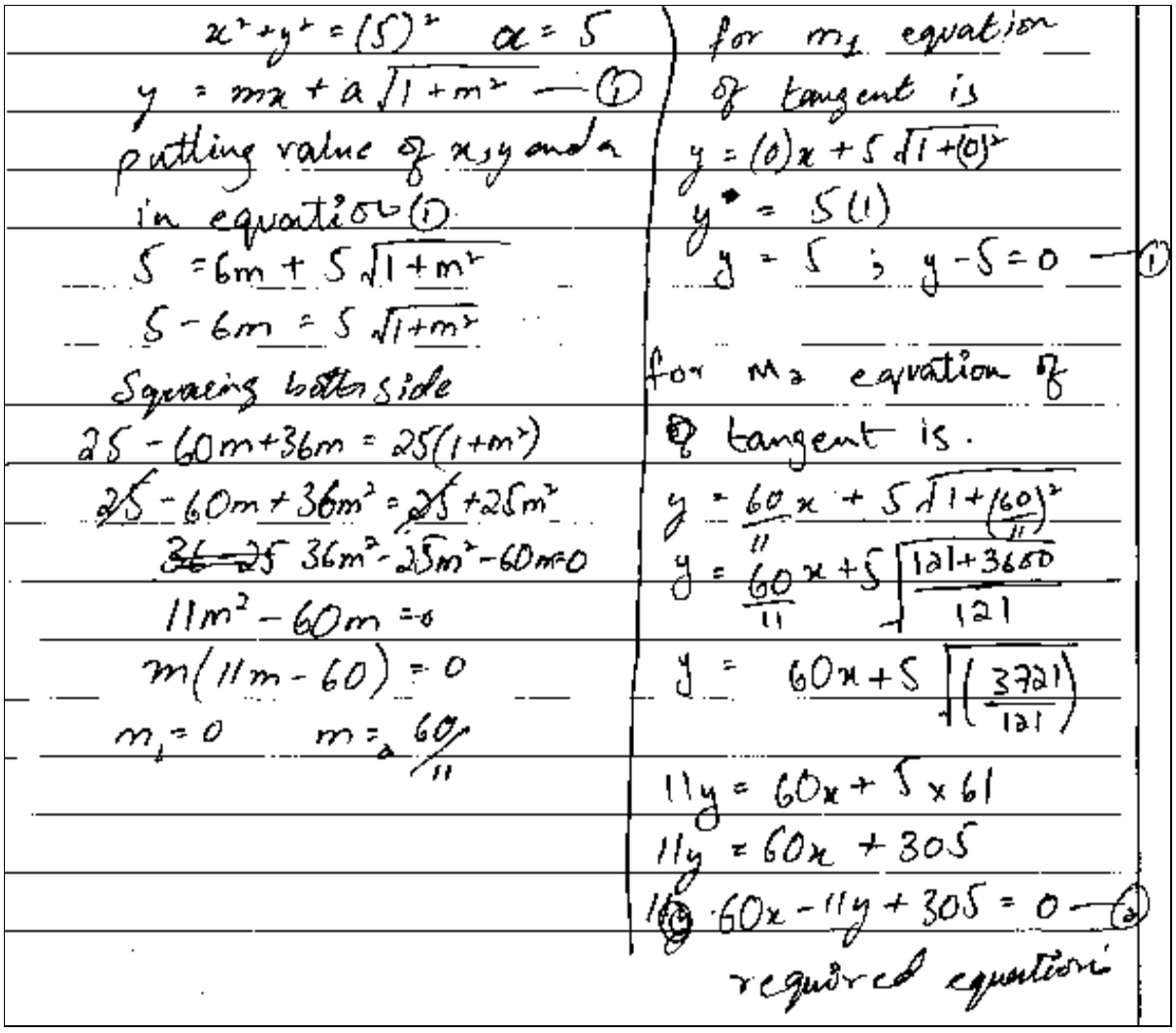
Question 6a:

Find the equations of two tangents from the point (6, 5) to the circle $x^2 + y^2 = 25$.

(**Note:** The equation of tangent from any point to the circle $x^2 + y^2 = a^2$ is given by $y = mx + a\sqrt{1+m^2}$, where m is the slope of the tangent.)

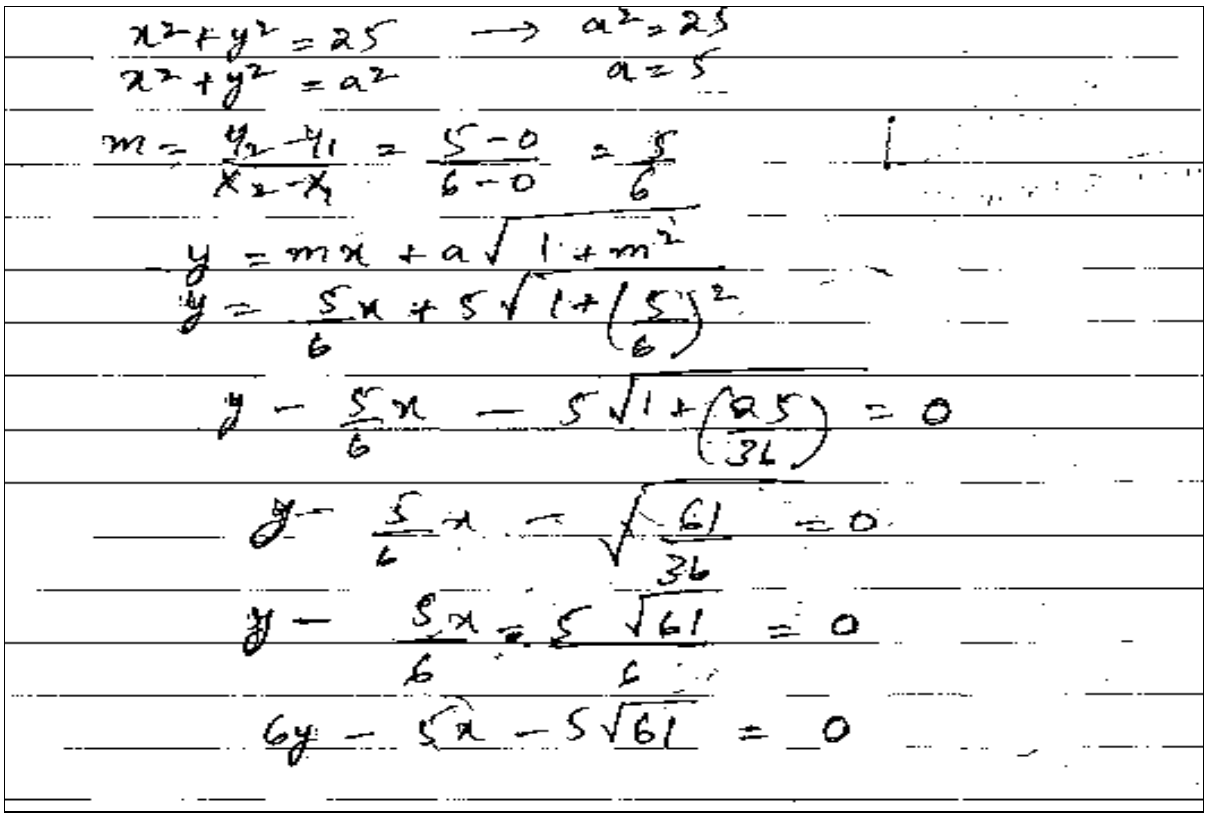
Better responses observed that candidates did well to understand the question and used the given formula accordingly. This helped candidates to go further in finding the gradients of tangent. As a result, they were able to get the desired answer.

Example:

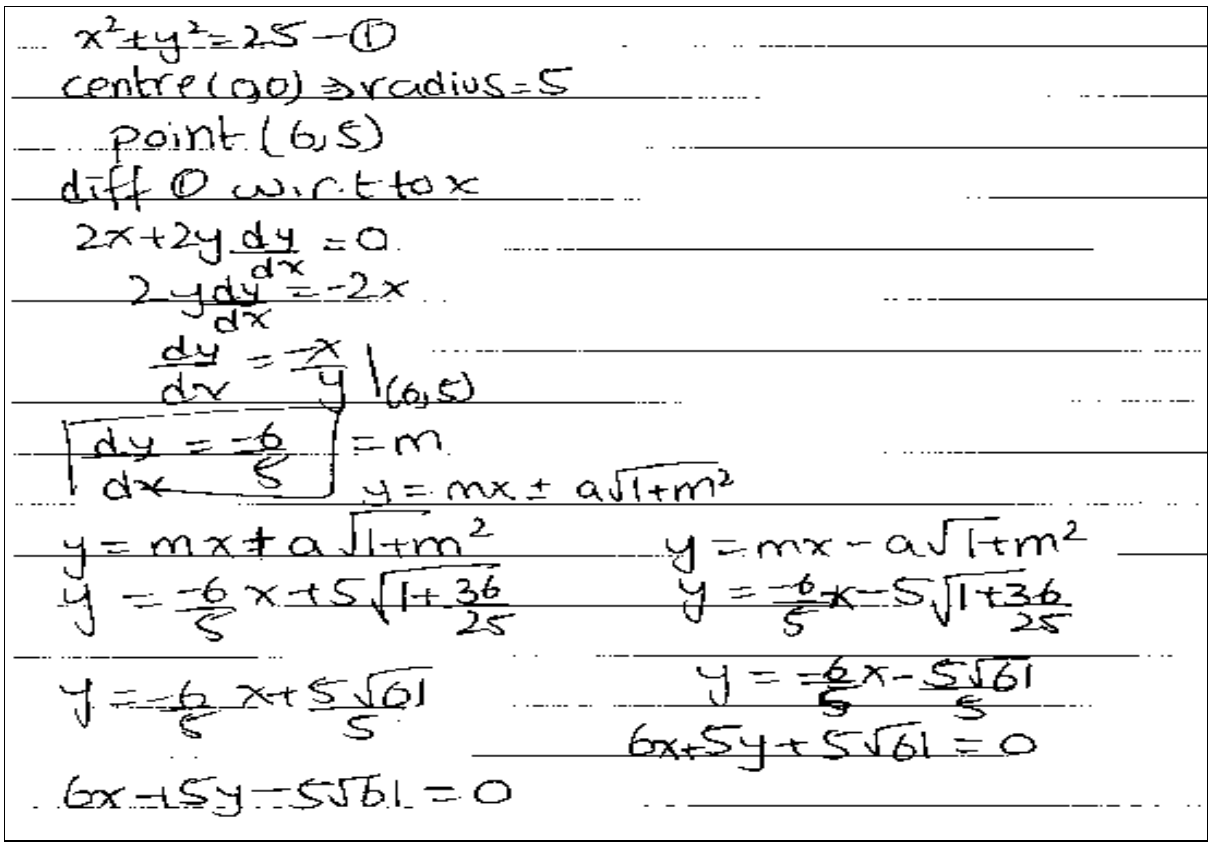


Weaker responses showed that candidates were able to understand parts of question and not the whole. They could not visualize the situation and gauged how the given data can be used to find the gradients of tangent. It was also noted that most of the candidates just managed to get one value of gradient rather than two. As a result, they were unable to achieve the desired answer. Multiple responses were observed as well even though method was incorrect.

Example 1:



Example 2:



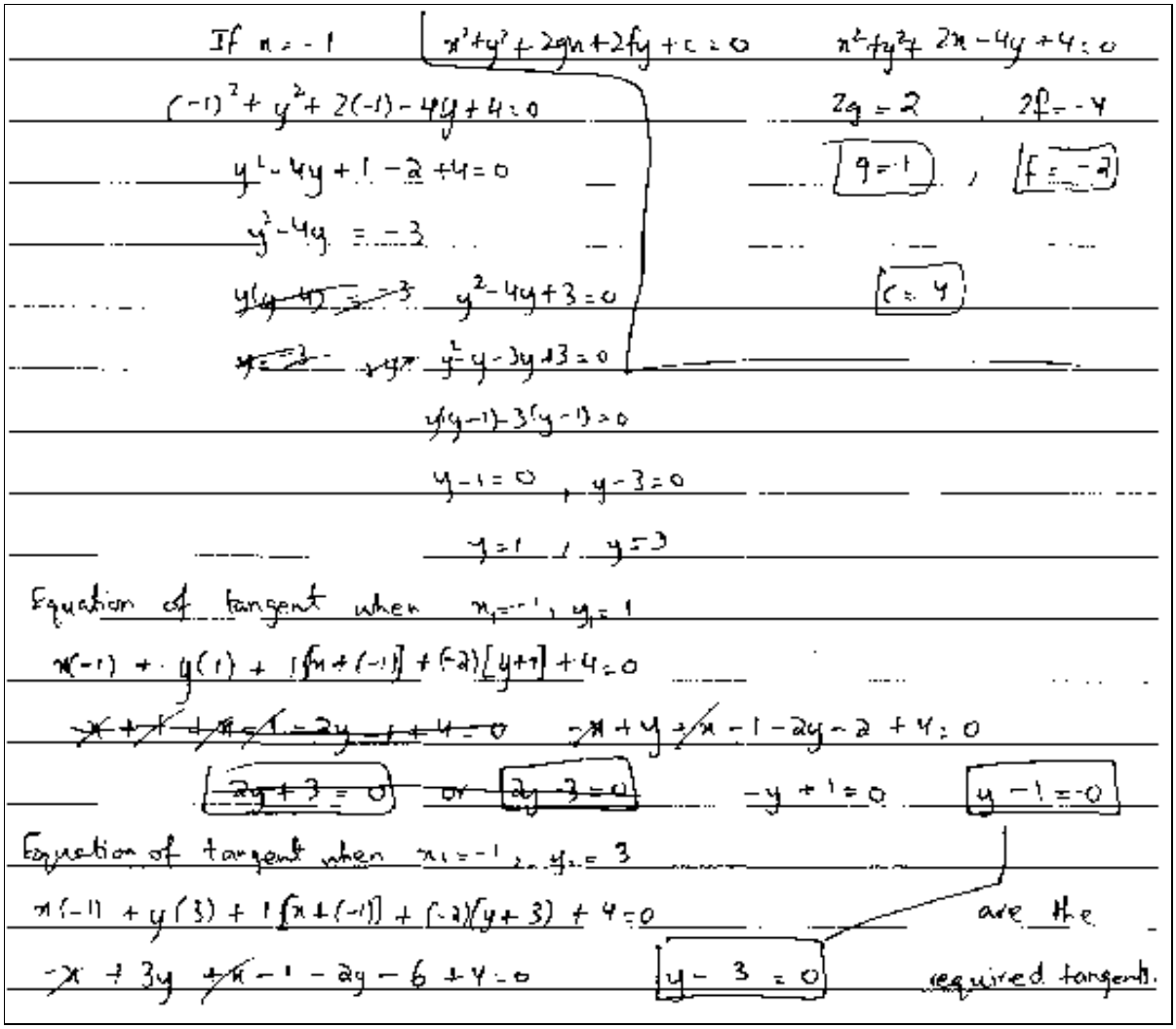
Question 6b:

Find the equations of tangents to the circle $x^2 + y^2 + 2x - 4y + 4 = 0$ at the points on the circle whose *x*-coordinate is − 1.

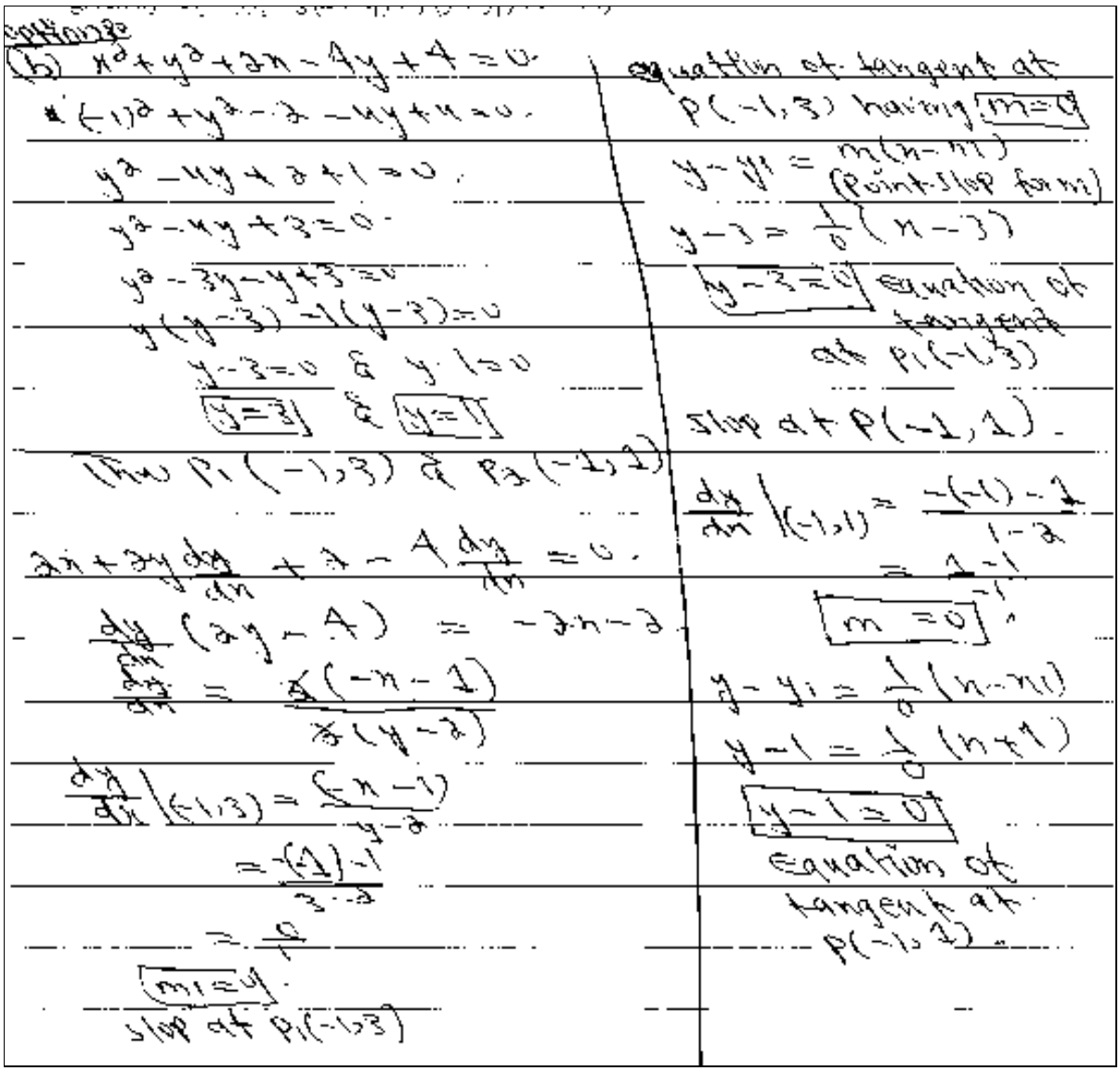
(**Note:** The equation of tangent at a point (x_1, y_1) on the circle $x_1^2 + y_1^2 + 2gx_1 + 2fy_1 + c = 0$ is given by $xx_1 + yy_1 + g(x + x_1) + f(y + y_1) + c = 0$.

Better responses reflected correct substitution of *x* – coordinate in the formula given in the question which motivated candidates to find *y*. By substituting these values, candidates were very much comfortable to proceed further in finding the required equations of tangent. Other responses were also noticed to achieve the answer.

Example 1:

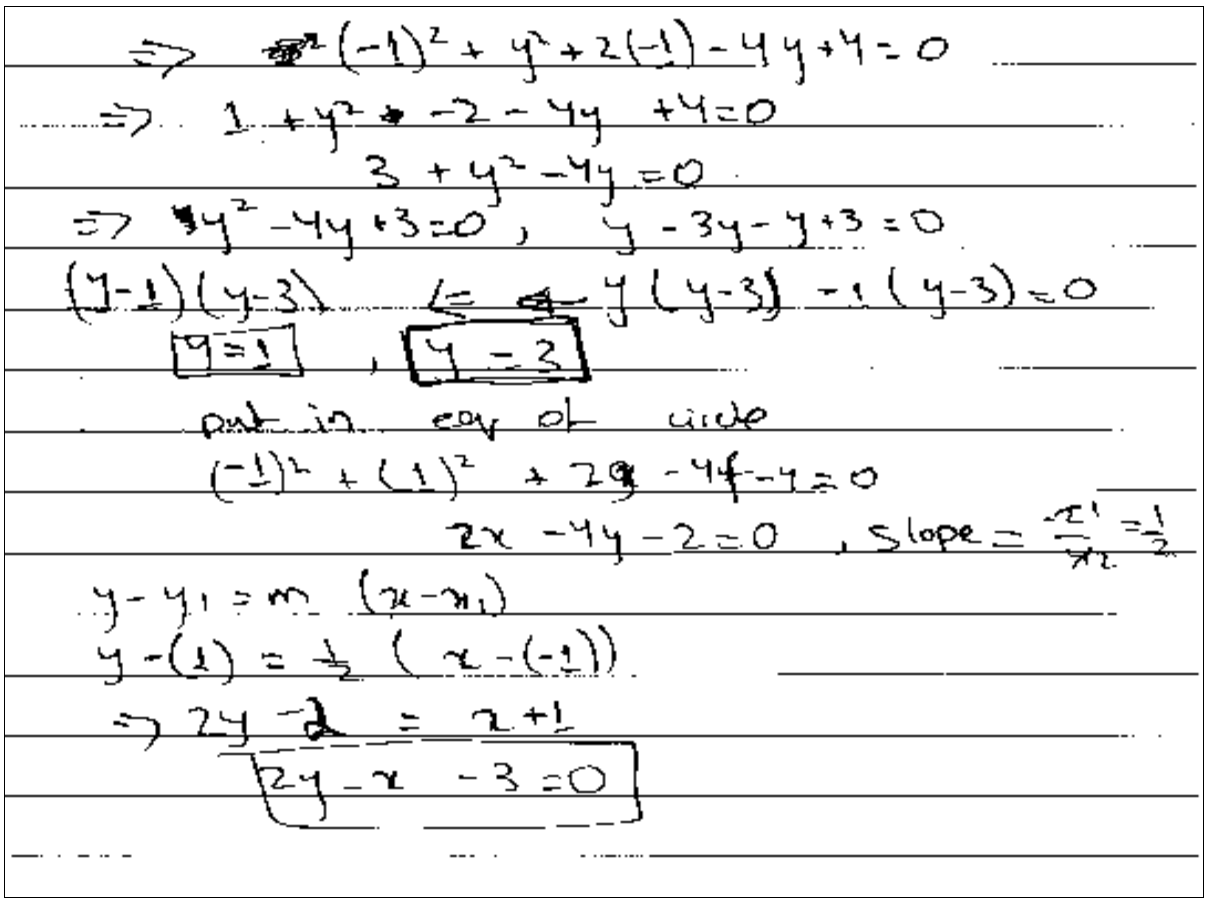


Example 2:



Weaker responses depicted vague understanding of question which showed lack of practice. They could not manage to get the right path in finding the answer. Fewer candidates although found the correct values of *y* but seemed, perplexed in finding the equations of tangent as they were directly using the equation of straight line $y - y_1 = m\left(x - x_1\right)$ without proceeding to any further steps.

Example:



Question 7:

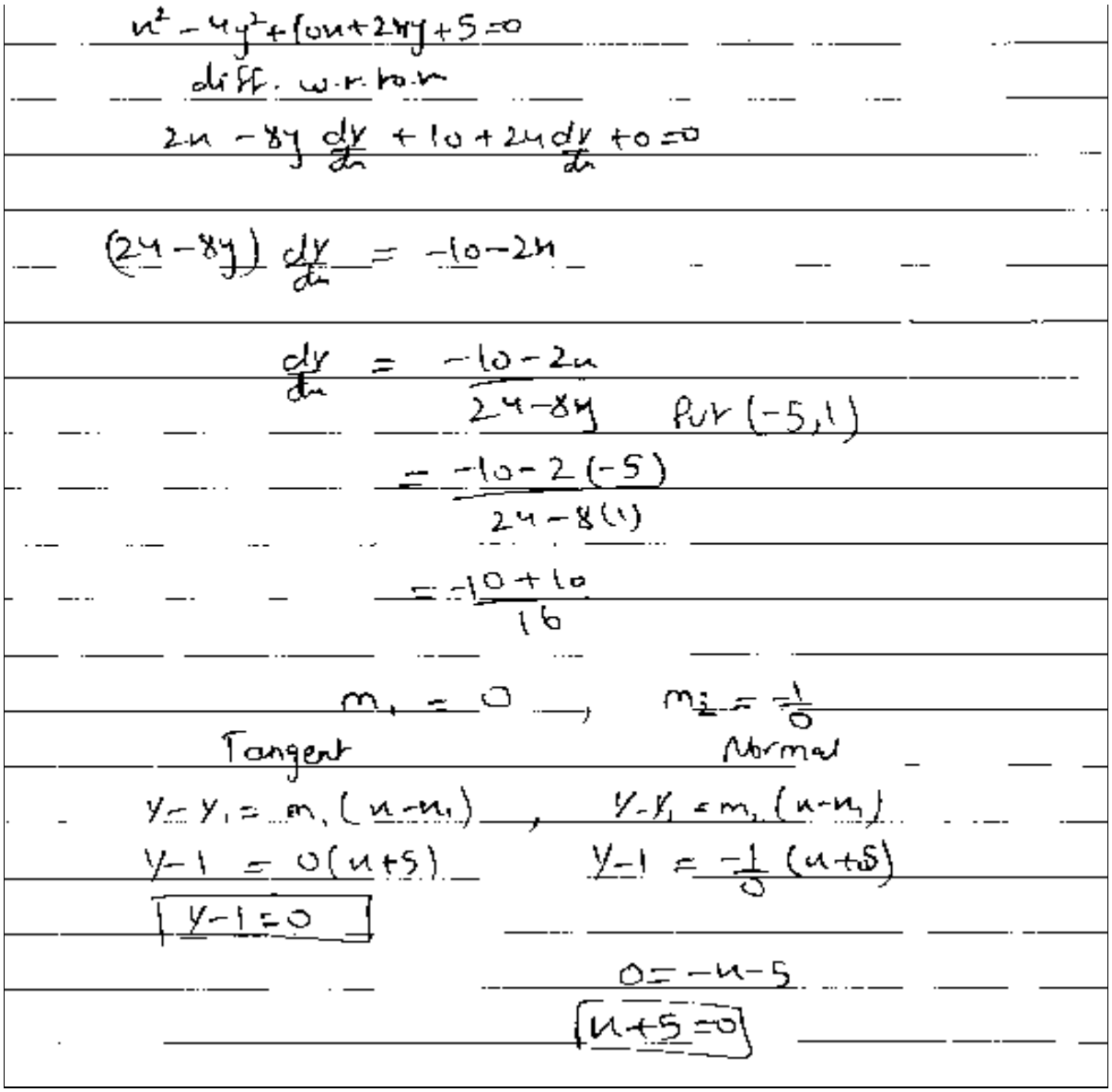
This question offered a choice to attempt any two questions from part **a**, **b** and **c**. Majority of the candidates attempted part **a** and part **b**. Fewer candidates attempted part **c** but did well.

Question 7a:

Find the equations of tangent and normal to the hyperbola $x^2 - 4y^2 + 10x + 24y + 5 = 0$ at (– 5, 1).

Better responses exhibited good understanding of tangent and normal to the hyperbola. The gradient of hyperbola at the given point was found correctly by using derivative. This directed candidates to get the right path in finding the equations of tangent and normal.

Example:



Weaker responses made mistakes in finding the derivative and got perplexed in finding the equation of tangent and normal. It was also seen that candidates although found the gradients correctly but ignored, the fact that $\frac{1}{0}$ cannot be changed anymore and wrote $\frac{1}{0}$ equals to 0 in finding the equation of normal which directed candidates to an incorrect answer.

Example:

$$\frac{dy}{dx} : 2x - 8y + 10 + 2y\frac{dy}{dx} = 0$$

$$dy/dx - 2\,dy/dx = 2x - 8y + 10$$

$$\frac{dy}{dx}[1-2] = 2x - 8y + 10$$

$$dy/dx = -2x + 8y - 10 \quad |(-5,1)$$

$$m = -2(-5) + 8(1) - 10$$

$$m = 10 + 8 - 10$$

$$\boxed{m = 8}$$

$m(x+5) = y-1$ for tangents

$$8(x+5) = y-1$$

$$8x + 40 = y - 1$$

$$\boxed{8x - y + 41 = 0}$$

for normal

$mm = -1$

$8m = -1$

$m = -\frac{1}{8}$

$$-\frac{1}{8}(x+5) = y-1$$

$$-\frac{1}{8}x - \frac{5}{8} = y - 1 \Rightarrow -x - 5 = 8y - 8$$

$$\Rightarrow 8y + x - 8 + 5 = 0$$

$$\Rightarrow \boxed{8y + x - 3 = 0}$$

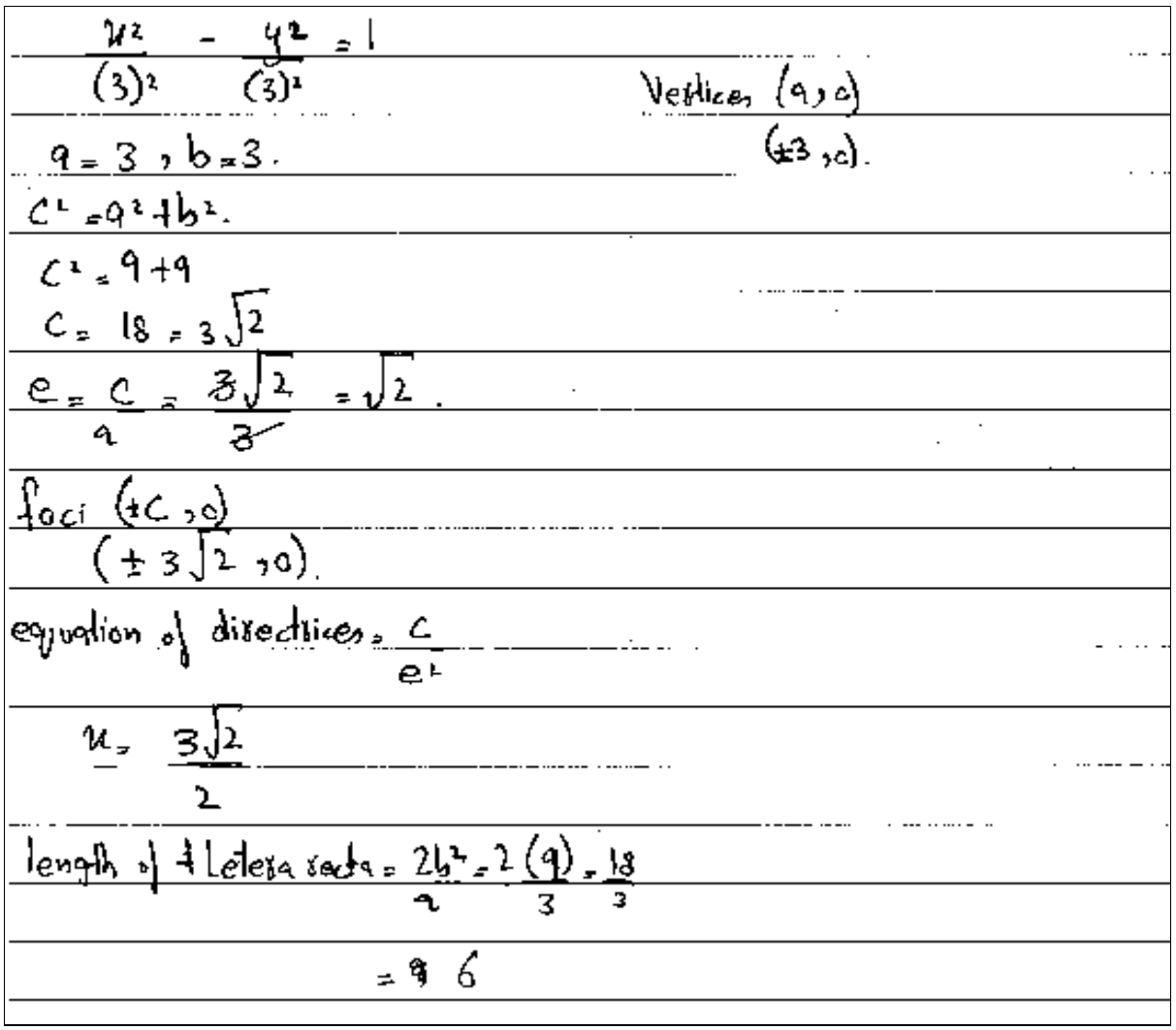
Question 7b:

For the hyperbola $x^2 - y^2 = 9$, find the

i. eccentricity.
ii. coordinates of foci.
iii. equations of directrices.
iv. lengths of latera recta.)
v. vertices.

Better responses showed that candidates used the equation of hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ to extract the data successfully in order to find a and b . This helped candidates in using the relationship $c^2 = a^2 + b^2$ correctly and therefore, they were able to get all the required values very easily.

Example:



Weaker responses depicted weak understanding of the elements of hyperbola. The candidates were not able to use the given information for their benefit. The main reason was their confusion about what *a*, *b* and *c* stand for. Therefore, they could not manage to achieve the desired answer.

One of the weaker responses is reflected in the example.

Example:

$x^2 - y^2 = 9$

Divide whole eq: by 9

$\frac{x^2}{9} - \frac{y^2}{9} = 1$

$c^2 = a^2 + b^2$

$c^2 = (3) + 3$

$c^2 = 6 \quad = \quad c = \pm\sqrt{6}$

i) $e = \pm \frac{c}{a} \Rightarrow \pm\frac{\sqrt{6}}{3}$

ii) Coordinates $= (0, \pm 3)$

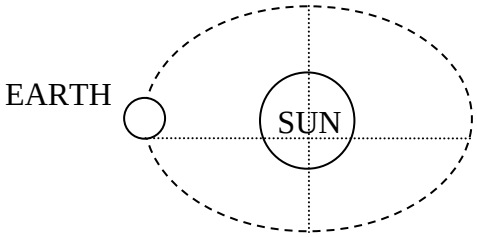
iii) $y = 3$

iv) $x = 3$

v) $(3, 0) \quad (-3, 0)$

Question 7c:

The earth is moving around the sun in an ELLIPTICAL path. During this motion, the shortest distance between their centres is 91 million miles while the farthest is 94.5 million miles.



If origin is considered as the centre of the sun, then the equation of the elliptical path will have the form $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$. Find the

i. values of *a* and *b*.

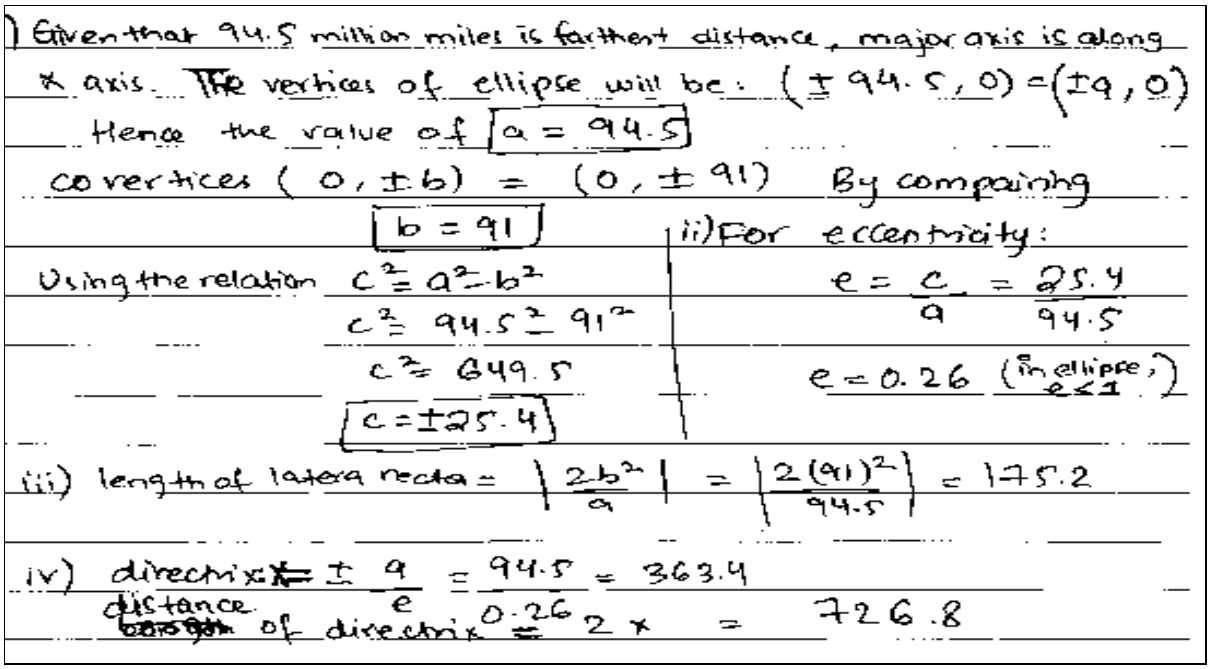
ii. eccentricity of this ellipse.

iii. lengths of latera recta.

iv. distance between directrices of the earth orbit.

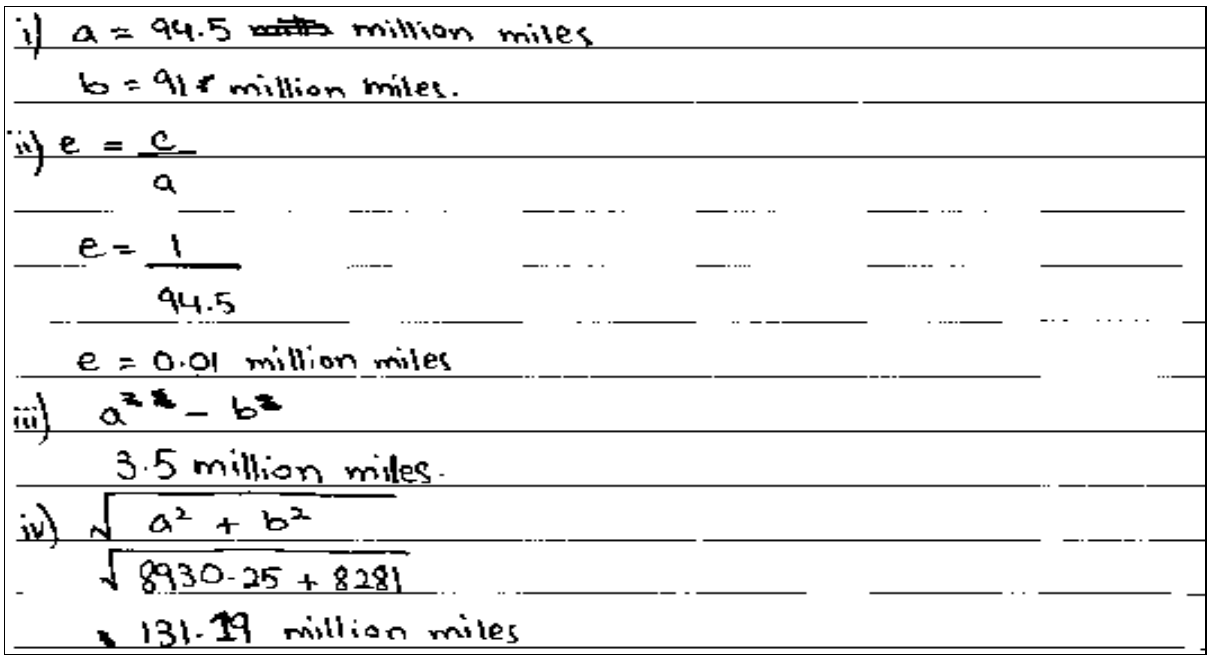
Better responses exhibited correct understanding of the elements of ellipse. The given data was used perfectly to find the values of *a*, *b* and *c*. This directed candidates to get the right path in finding the different elements of ellipse.

Example:


i) Given that 94.5 million miles is farthest distance, major axis is along x axis. The vertices of ellipse will be: $(\pm 94.5, 0) = (\pm a, 0)$
Hence the value of $\boxed{a = 94.5}$
co vertices $(0, \pm b) = (0, \pm 91)$ By comparing
$\boxed{b = 91}$
Using the relation $c^2 = a^2 - b^2$
$c^2 = 94.5^2 - 91^2$
$c^2 = 649.5$
$\boxed{c = \pm 25.4}$
ii) For eccentricity:
$e = \frac{c}{a} = \frac{25.4}{94.5}$
$e = 0.26$ (in ellipse, $e < 1$)
iii) length of latera recta $= \left|\frac{2b^2}{a}\right| = \left|\frac{2(91)^2}{94.5}\right| = 175.2$
iv) directrix $= \pm \frac{a}{e} = \frac{94.5}{0.26} = 363.4$
distance of directrix $= 2x = 726.8$

Weaker responses failed to grasp the question. They could not translate the given information in the question into an equation. It was also seen that some candidates managed to find the values of *a*, *b* and *c* but, could not get the required elements due to lack of practice.

Example:



Question 8:

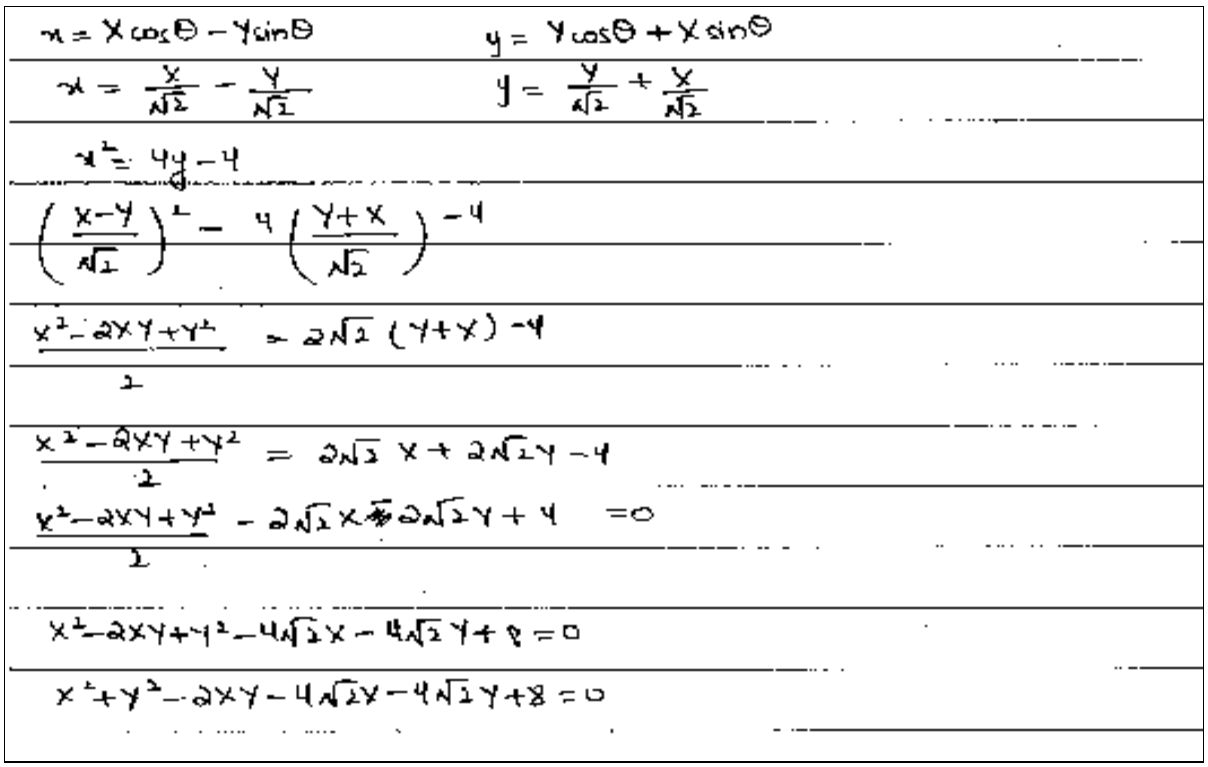
This question offered a choice between part **a** and part **b**. Majority of the candidates chose part **a** as compared to part b which was also attempted much better than part **b**. Although it was a repeated question from past papers, it was not well attempted.

Question 8a:

Find the equation of the parabola $x^2 = 4y - 4$ with respect to new axes obtained by rotation of axes about the origin through an angle of $45°$.

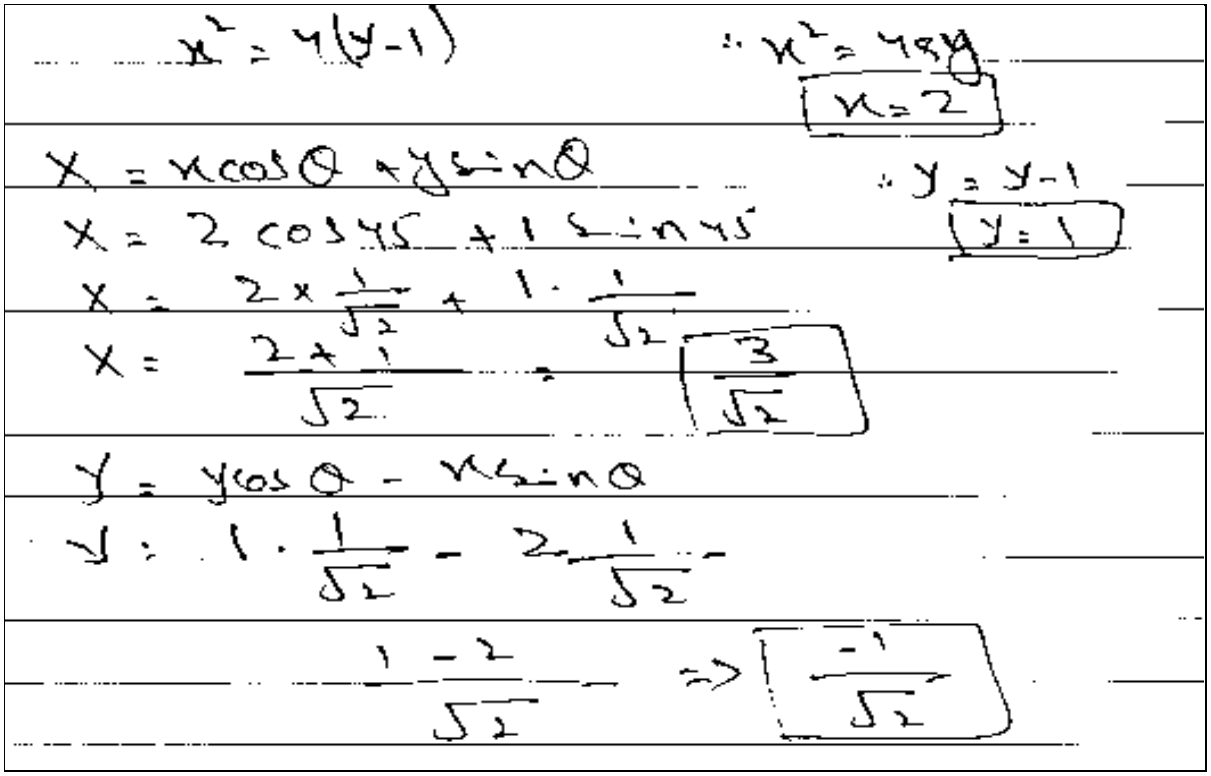
Better responses indicated the correct use of equations of transformation to get equations in terms of X and Y i.e., $x = X\cos\theta - Y\sin\theta$ and $y = X\sin\theta + Y\cos\theta$. This helped candidates by substituting these values in the given equation in the right path to get the required answer.

Example:



Weaker responses reflected that candidates mostly used incorrect equations of substitution. The most frequent incorrect equations used were $X = x\cos\theta - y\sin\theta$ and $Y = y\cos\theta + x\sin\theta$ which led the candidates to incorrect answers.

Example:

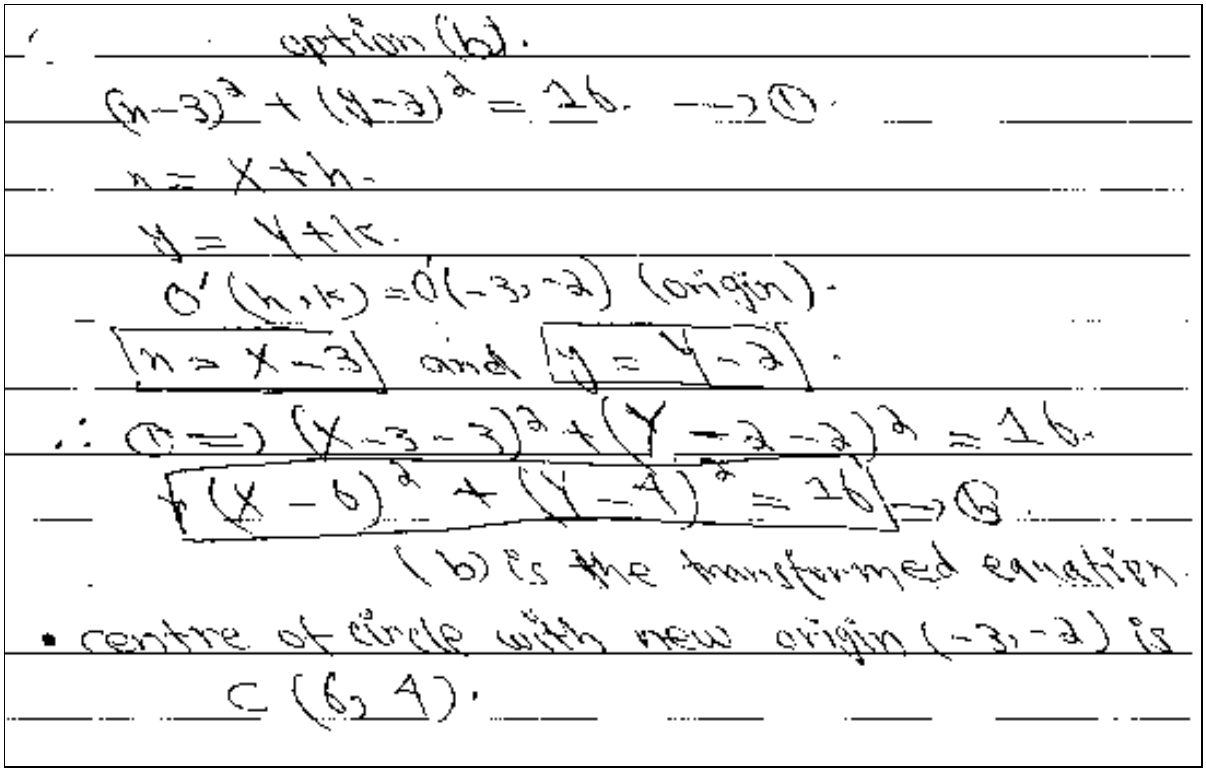


Question 8b:

A circle is defined as $(x-3)^2+(y-2)^2=16$ with reference to *xy*-coordinate axes. Transform the equation of this circle with reference to (– 3, – 2) as new origin, such that the new axes remain parallel to the old axes. Also write the centre of circle with reference to the new origin (– 3, – 2).

Better responses indicated that candidates used the correct transformation which directed to the correct answer. They substituted $x = X - 3$ and $y = Y - 2$ in the given equation and as a result, they were able to reach the required answer.

Example:



Weaker responses observed that candidates did not understand the question. It was seen that they were confused in using the transformation and made mistakes in writing as they wrote $x = X + 3$ and $y = Y + 2$ which was actually incorrect. Because of this, they were unable to get the required answer.

Example:

